

A MAJOR PROJECT REPORT
ON
“SYSTEM FOR IDENTIFYING FAKE PRODUCT USING
BLOCKCHAIN TECHNOLOGY”

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SRI INDU COLLEGE OF ENGINEERING & TECHNOLOGY

(An Autonomous Institution under UGC, Accredited by NBA, Affiliated to JNTUH)

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DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY



CERTIFICATE

Certified that the Major Project entitled “**SYSTEM FOR IDENTIFYING FAKE PRODUCT USING BLOCKCHAIN TECHNOLOGY**” is a bonafide work carried out by **K.AKSHITHA [21D41A3336], Y.SWETHA [21D41A3364], K.KARTHIK [21D41A3334], CH.KARTHIKEYA [21D41A3308]** in partial fulfillment for the award of **Bachelor of Technology in Computer Science and Information Technology** of SICET, Hyderabad for the academic year 2024- 2025. The Project has been approved as it satisfies academic requirements in respect of the work prescribed for **IV YEAR, II-SEMESTER** of **B. TECH** course.

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SYSTEM FOR IDENTIFYING FAKE PRODUCT USING BLOCKCHAIN TECHNOLOGY

ABSTRACT

Counterfeit items have become more prevalent in the industrial industry in recent years. This has an effect on the company's name, sales, and profit. Blockchain technology is used to authenticate and identify counterfeit items. Using blockchain technology, transactions are recorded in the form of blocks in a distributed, decentralized, and digital ledger. A block cannot be changed or hacked since blockchain technology is secure. Blockchain technology eliminates the need for customers or users to rely on third parties to verify the product's safety. In light of current advancements in mobile and wireless technologies, quick response (QR) codes provide a promising technique for addressing the practice of counterfeiting in this project. This is done by scanning a QR code, which is linked to a Blockchain, to identify counterfeit goods. As a consequence, this method may be used to record product information and generate a unique identifier for each product in a database. It looks up the user's unique code in the Blockchain database and sees if there are any matches. When a code matches, the client will be notified; otherwise, the consumer will be notified that the product is counterfeit.

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1. INTRODUCTION

1.1 INTRODUCTION

The company's brand, revenue, and consumer health might be at jeopardy if a product or technology is made available to a global audience. The supply chain contains an endless number of things. In order to assess whether or not a product is authentic or a fake. As manufacturers are experiencing a major challenge so we can utilize blockchain technology to verify the authenticity of a product [1-3]. Changing, hacking, or defrauding the system is difficult or impossible using Blockchain's data recording technique. In a blockchain, transactions are duplicated and propagated to all of the network's PC computers in the form of a record of transactions. Decentralized databases are known as Distributed Ledger Technology (DLT) and are maintained by huge groups of people (DLT). Blockchain is a distributed ledger system that uses an immutable cryptographic signature known as a hash to record transactions [4-5]. The blockchain will keep all transaction data in the form of blocks. A QR code generated for a particular product may be scanned by the end user to get further information about that product. We can tell if a thing is genuine or not by scanning its QR code [6].

A. MOTIVATION

There The distribution of counterfeit products has grown widespread in recent years. There are several counterfeit items in today's supply chain. According to the research, occurrences involving bogus products have increased in recent years. It is vital to have a system in place for consumers or users to verify all product facts in order for users to determine if the product is genuine or not. Currently, India lacks such a mechanism for detecting counterfeit items [7-8].

B. OBJECTIVE

The growth in counterfeit goods sparked the idea for this project. Following are the project's goals: 1. An anti-counterfeit system based on blockchain technology is being developed. 2. To use a QR code to encrypt product information. 3. Customers' security is ensured by supplying them with their own data.

1.2 LITERATURE SURVEY

A Survey on Counterfeit Product Detection. Counterfeit items are exploding in popularity, thanks to the vast internet and black-market marketplaces. As a result, there is an urgent need to solve the difficulties associated with identifying counterfeit items and to develop suitable technologies to increase detection accuracy. This is an active field of study in the modern world. This article explores numerous approaches for determining whether or not a product is counterfeit [9-10]. A framework for supply chain quality management based on the blockchain. In this work, we suggest a blockchain-based system. The theoretical basis for blockchain-based supply chain quality management will be laid forth in this framework [11]. Additionally, it serves as a basis for the development of ideas regarding the management of information resources in dispersed, virtual companies [12-15].

1.3 MODULES

Product Registration and Blockchain Integration:

In this module, manufacturers or authorized entities register genuine products on the system. Each product is assigned a unique identifier, and relevant information, such as production details, is recorded on the blockchain. Integration with the blockchain ensures the secure and transparent storage of product-related transactions.

QR Code Generation and Linking:

This module involves the generation and linking of QR codes to each registered product. A QR code is a visual representation of the unique identifier associated with a specific product. These QR codes serve as a bridge between the physical product and the blockchain, enabling consumers to easily access product information by scanning the code.

Consumer Interaction and Verification:

This module facilitates consumer interaction with the system. Consumers use their smartphones or dedicated scanning devices to scan the QR code on a product. The system then initiates a verification process by querying the blockchain database using the unique identifier. If a match is found, the product is deemed authentic; otherwise, it is flagged as potentially counterfeit.

Blockchain Database Management:

The blockchain database management module is responsible for maintaining and updating the distributed ledger. It ensures the integrity of recorded transactions and manages access permissions to safeguard sensitive information. This module plays a critical role in maintaining the security and transparency of the system.

Notification and Reporting:

This module handles the communication aspect of the system. When a consumer scans a product, the system generates real-time notifications based on the verification results. If the product is authentic, the consumer is notified of its legitimacy. In the case of a potential counterfeit, the system alerts the consumer and may trigger further reporting mechanisms to track and address the issue.

2.SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

Several initiatives and projects around the world have explored or implemented blockchain-based solutions for supply chain management and anti-counterfeiting. These systems often involve a combination of blockchain, QR codes, and databases to provide transparency and traceability in the supply chain. Companies, particularly in industries where counterfeiting is a significant concern (such as pharmaceuticals, luxury goods, and electronics), have been exploring the use of blockchain to enhance product authentication. For the most accurate and up-to-date information on existing systems, I recommend checking recent academic publications, industry news, and official project websites. Additionally, contacting experts in the field of blockchain technology and supply chain management could provide valuable insights.

LIMITATIONS

Adoption Challenges:

One major limitation is the challenge of widespread adoption. For the system to be effective, manufacturers, distributors, and retailers all need to participate. If key players in the supply chain do not adopt the technology, it becomes challenging to create a comprehensive and reliable record on the blockchain.

Data Accuracy and Trust:

The system's effectiveness heavily relies on the accuracy of the data entered into the blockchain. If inaccurate information is recorded at any point in the supply chain, it can compromise the integrity of the entire system. Building trust in the data recorded is crucial for the success of the anti-counterfeiting measures.

Technology Integration:

Integrating blockchain technology into existing supply chain systems can be complex and may require significant changes. Some businesses may face challenges in adapting their current processes and technologies to seamlessly work with the blockchain-based system.

Privacy Concerns:

While blockchain is known for its security, there are still privacy concerns related to the information stored on the blockchain. Ensuring that sensitive information is appropriately protected while maintaining transparency is a delicate balance. Striking the right balance between transparency and privacy is crucial.

Scalability:

Scalability can be a concern as the volume of transactions and participants in the supply chain grows. Blockchain networks may face challenges in handling a large number of transactions simultaneously, leading to potential delays and increased costs.

2.2 PROPOSED SYSTEM

The proposed system aims to revolutionize the battle against counterfeit products by leveraging the security and transparency features of blockchain technology. In our system, each product is assigned a unique identifier linked to a blockchain, creating an immutable and decentralized ledger of transactions. The heart of our solution lies in the integration of Quick Response (QR) codes, providing a user-friendly interface for consumers to easily interact with the blockchain. Upon scanning the QR code, users can access the blockchain database to verify the authenticity of the product. This process involves a lookup of the unique identifier, crossreferencing it with the information stored in the blockchain. In the context of the ever-evolving mobile and wireless technologies, QR codes present a promising tool for combating counterfeiting. The system not only records essential product information but also generates a distinct code for each item, enhancing traceability and accountability throughout the supply chain.

One of the notable advantages of our proposed system is the elimination of reliance on thirdparty verification. By directly connecting consumers with the blockchain, we empower them to independently verify the safety and authenticity of a product. This approach not only enhances consumer trust but also streamlines the verification process, reducing the need for intermediaries. Furthermore, the immutability of blockchain ensures that once a transaction is recorded, it remains incorruptible, providing a robust defense against tampering or hacking.

ADVANTAGES

Enhanced Security and Immutability:

Blockchain's inherent security features ensure that once a transaction is recorded, it cannot be altered or tampered with. This immutability provides a robust defense against counterfeiting activities, offering a secure and transparent record of product transactions.

Direct Consumer Empowerment:

The system empowers consumers by enabling them to independently verify the authenticity of a product. Through the use of QR codes and blockchain technology, consumers can directly access information, eliminating the need for third-party verification. This fosters trust and confidence among consumers in the products they purchase.

Streamlined Supply Chain Transparency:

By integrating blockchain into the supply chain, the proposed system enhances transparency and traceability. Each product is assigned a unique identifier, and its journey through the supply chain is recorded on the blockchain. This streamlined transparency reduces the likelihood of counterfeit products entering the market and allows for quick identification of any irregularities.

Reduction in Counterfeiting and Fraud:

The system acts as a deterrent to counterfeiters by creating a secure and traceable record of genuine products. The immutability of blockchain makes it difficult for counterfeiters to replicate or manipulate product information, reducing the prevalence of counterfeit goods in the market.

Efficient and Decentralized Verification:

The use of QR codes and blockchain technology allows for efficient and decentralized product verification. Consumers can quickly scan a QR code using their smartphones, initiating a direct connection to the blockchain database. This decentralized approach eliminates the need for centralized verification authorities, making the verification process more efficient and accessible.

2.3 SYSTEM REQUIREMENTS

SOFTWARE REQUIREMENTS	
Operating System:	Windows 10/11
Development Software:	Python 3.10
Programming Language:	Python
Integrated Development Environment (IDE):	Visual Studio Code
Front End Technologies:	HTML5, CSS3, Java Script
Back End Technologies or Framework:	Django
Database Language:	SQL
Database (RDBMS):	MySQL
Web server or deployment server:	Django Application server

HARDWARE REQUIREMENTS	
MY SYSTEM (Development) System	
Processor:	intel core i5 Processor 8th Gen
Hard Disk:	256 Gb
RAM:	8Gb

3. SYSTEM STUDY

3.1 FEASIBILITY STUDY

A feasibility study assesses the operational, technical and economic merits of the proposed project. The feasibility study is intended to be a preliminary review of the facts to see if it is worthy of proceeding to the analysis phase. From the systems analyst perspective, the feasibility analysis is the primary tool for recommending whether to proceed to the next phase or to discontinue the project.

The feasibility study is a management-oriented activity. The objective of a feasibility study is to find out if an information system project can be done and to suggest possible alternative solutions.

Three key considerations involved in the feasibility analysis are

- ECONOMICAL FEASIBILITY
- TECHNICAL FEASIBILITY
- SOCIAL FEASIBILITY

Projects are initiated for two broad reasons:

1. Problems that lend themselves to systems solutions
2. Opportunities for improving through:
 - (a) upgrading systems
 - (b) altering systems
 - (c) installing new systems

A feasibility study should provide management with enough information to decide:

- Whether the project can be done
- Whether the final product will benefit its intended users and organization
- What are the alternatives among which a solution will be chosen?
- Is there a preferred alternative?

TECHNICAL FEASIBILITY

A large part of determining resources has to do with assessing technical feasibility. It considers the technical requirements of the proposed project. The technical requirements are then compared to the technical capability of the organization. The systems project is considered technically feasible if the internal technical capability is sufficient to support the project requirements.

The analyst must find out whether current technical resources can be upgraded or added to in a manner that fulfils the request under consideration. This is where the expertise of system analysts is beneficial, since using their own experience and their contact with vendors they will be able to answer the question of technical feasibility.

The essential questions that help in testing the operational feasibility of a system include the following:

- Is the project feasible within the limits of current technology?
- Does the technology exist at all?
- Is it available within given resource constraints?
- Is it a practical proposition?
- Manpower- programmers, testers & debuggers
- Software and hardware
- Are the current technical resources sufficient for the new system?
- Can they be upgraded to provide to provide the level of technology necessary for the new system?
- Do we possess the necessary technical expertise, and is the schedule reasonable?
- Can the technology be easily applied to current problems?
- Does the technology have the capacity to handle the solution?
- Do we currently possess the necessary technology?

OPERATIONAL FEASIBILITY

Operational feasibility is dependent on human resources available for the project and involves projecting whether the system will be used if it is developed and implemented. Operational feasibility is a measure of how well a proposed system solves the problems, and takes advantage of the opportunities identified during scope definition and how it satisfies the requirements identified in the requirements analysis phase of system development.

Operational feasibility reviews the willingness of the organization to support the proposed system. This is probably the most difficult of the feasibilities to gauge. In order to determine this feasibility, it is important to understand the management commitment to the proposed project. If the request was initiated by management, it is likely that there is management support and the system will be accepted and used. However, it is also important that the employee base will be accepting of the change.

The essential questions that help in testing the operational feasibility of a system include the following:

- Does current mode of operation provide adequate throughput and response time?
- Does current mode provide end users and managers with timely, pertinent, accurate and useful formatted information?
- Does current mode of operation provide cost-effective information services to the business?
- Could there be a reduction in cost and or an increase in benefits?
- Does current mode of operation offer effective controls to protect against fraud and to guarantee accuracy and security of data and information?
- Does current mode of operation make maximum use of available resources, including people, time, and flow of forms?
- Does current mode of operation provide reliable services
- Are the services flexible and expandable?
- Are the current work practices and procedures adequate to support the new system?
- If the system is developed, will it be used?

- Manpower problems
- Labor objections
- Manager resistance
- Organizational conflicts and policies
- Social acceptability
- Government regulations
- Does management support the project?
- Are the users not happy with current business practices?
- Will it reduce the time (operation) considerably?
- Have the users been involved in the planning and development of the project?
- Will the proposed system really benefit the organization?
- Does the overall response increase?
- Will accessibility of information be lost?
- Will the system affect the customers in considerable way?
- Legal aspects
- How do the end-users feel about their role in the new system?
- What end-users or managers may resist or not use the system?
- How will the working environment of the end-user change?
- Can or will end-users and management adapt to the change?

ECONOMIC FEASIBILITY

Economic analysis could also be referred to as cost/benefit analysis. It is the most frequently used method for evaluating the effectiveness of a new system. In economic analysis the procedure is to determine the benefits and savings that are expected from a candidate system and compare them with costs. If benefits outweigh costs, then the decision is made to design and implement the system. An entrepreneur must accurately weigh the cost versus benefits before taking an action.

Possible questions raised in economic analysis are:

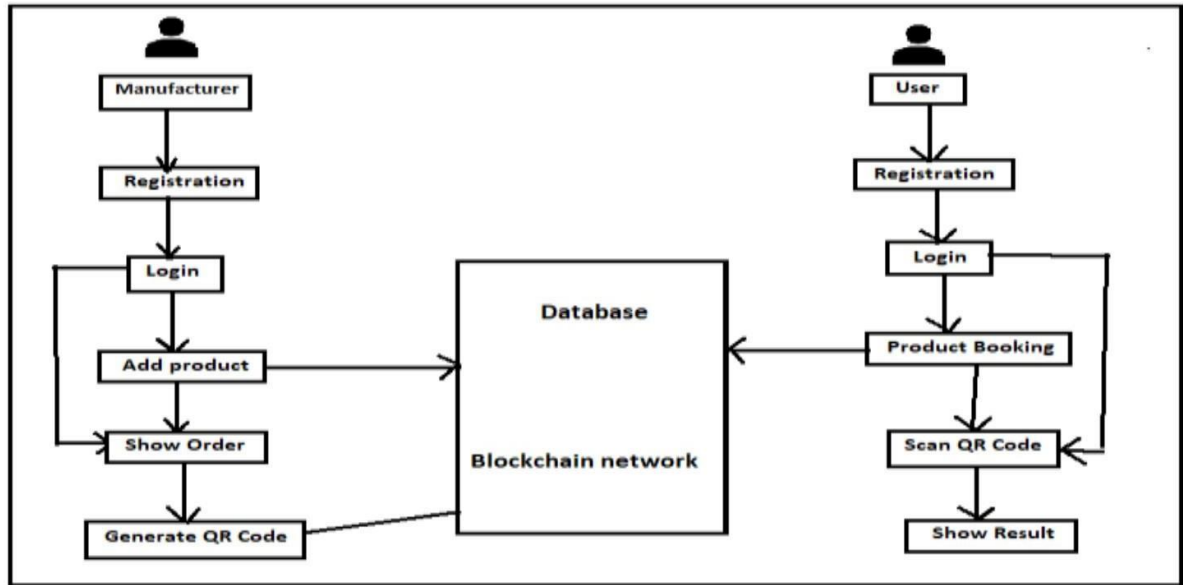
- Is the system cost effective?
- Do benefits outweigh costs?
- The cost of doing full system study
- The cost of business employee time
- Estimated cost of hardware
- Estimated cost of software/software development • Is the project possible, given the resource constraints?
- What are the savings that will result from the system?
- Cost of employees' time for study
- Cost of packaged software/software development
- Selection among alternative financing arrangements (rent/lease/purchase)

The concerned business must be able to see the value of the investment it is pondering before committing to an entire system study. If short-term costs are not overshadowed by long-term gains or produce no immediate reduction in operating costs, then the system is not economically feasible, and the project should not proceed any further. If the expected benefits equal or exceed costs, the system can be judged to be economically feasible. Economic analysis is used for evaluating the effectiveness of the proposed system.

The economic feasibility will review the expected costs to see if they are in-line with the projected budget or if the project has an acceptable return on investment. At this point, the projected costs will only be a rough estimate. The exact costs are not required to determine economic feasibility. It is only required to determine if it is feasible that the project costs will fall within the target budget or return on investment. A rough estimate of the project schedule is required to determine if it would be feasible to complete the systems project within a required timeframe. The required timeframe would need to be set by the organization.

4. SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE



4.2 UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of objectoriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects-oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

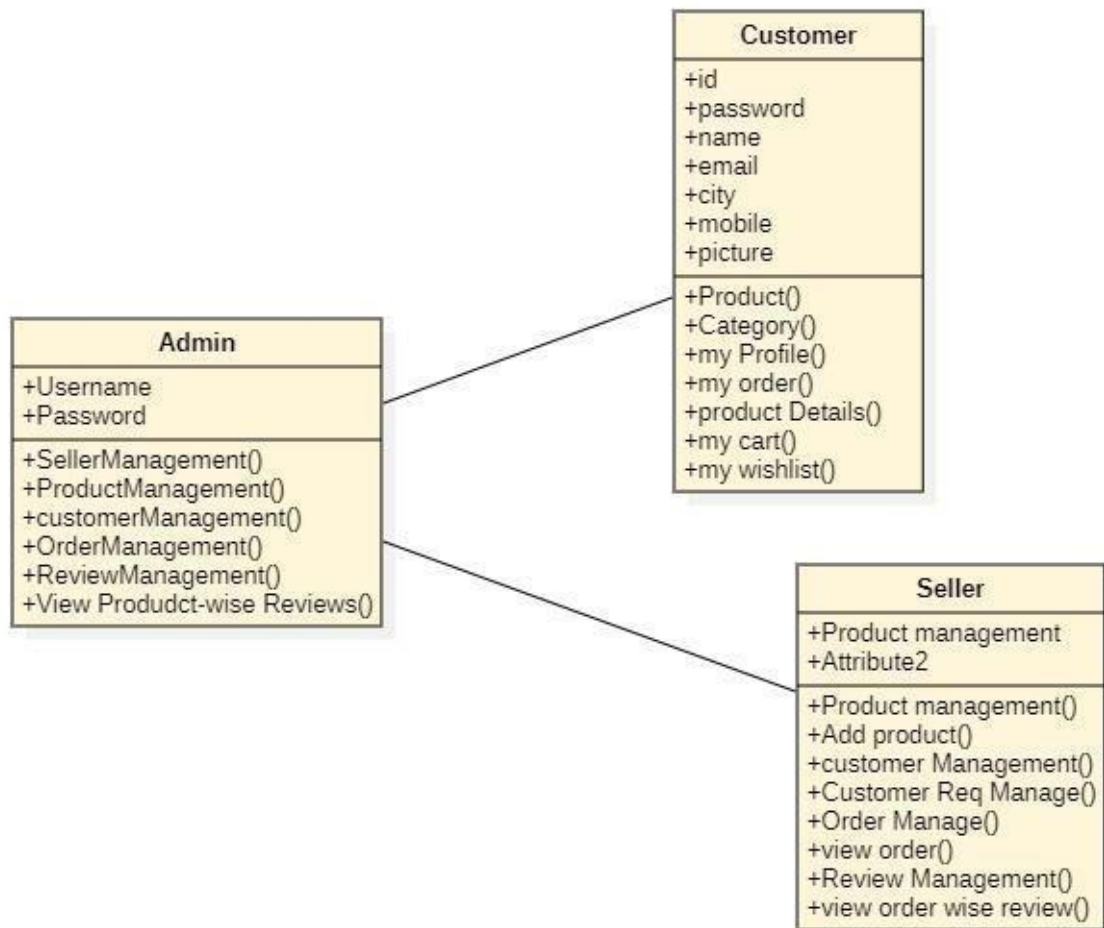
GOALS:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Encourage the growth of OO tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns and components.
7. Integrate best practices.

4.2.1 CLASS DIAGRAM

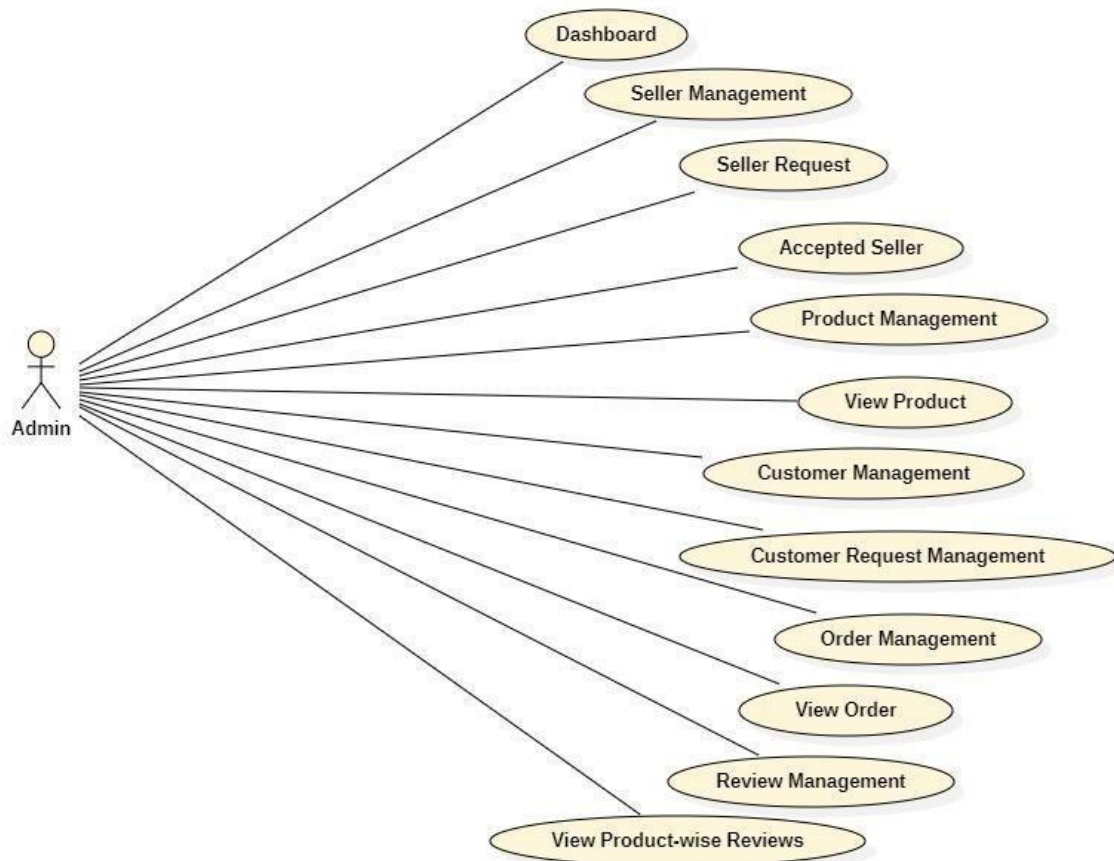
In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.



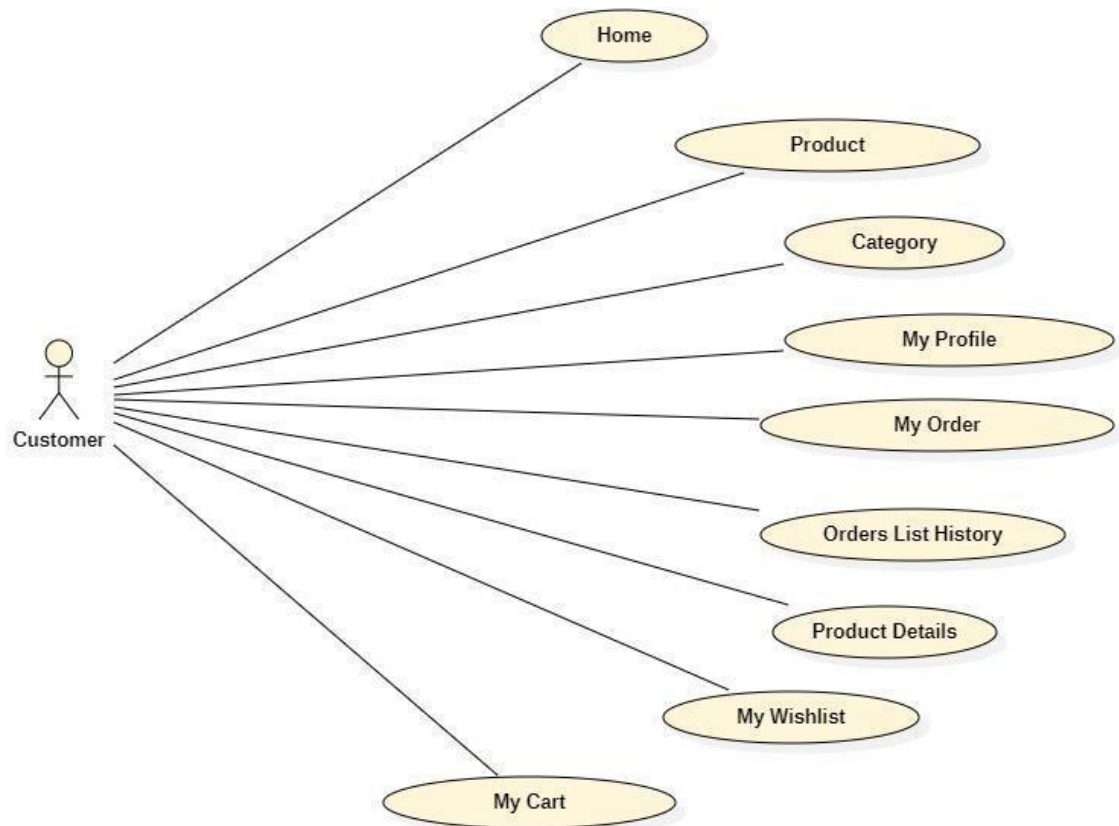
4.2.2 USE CASE DIAGRAM

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

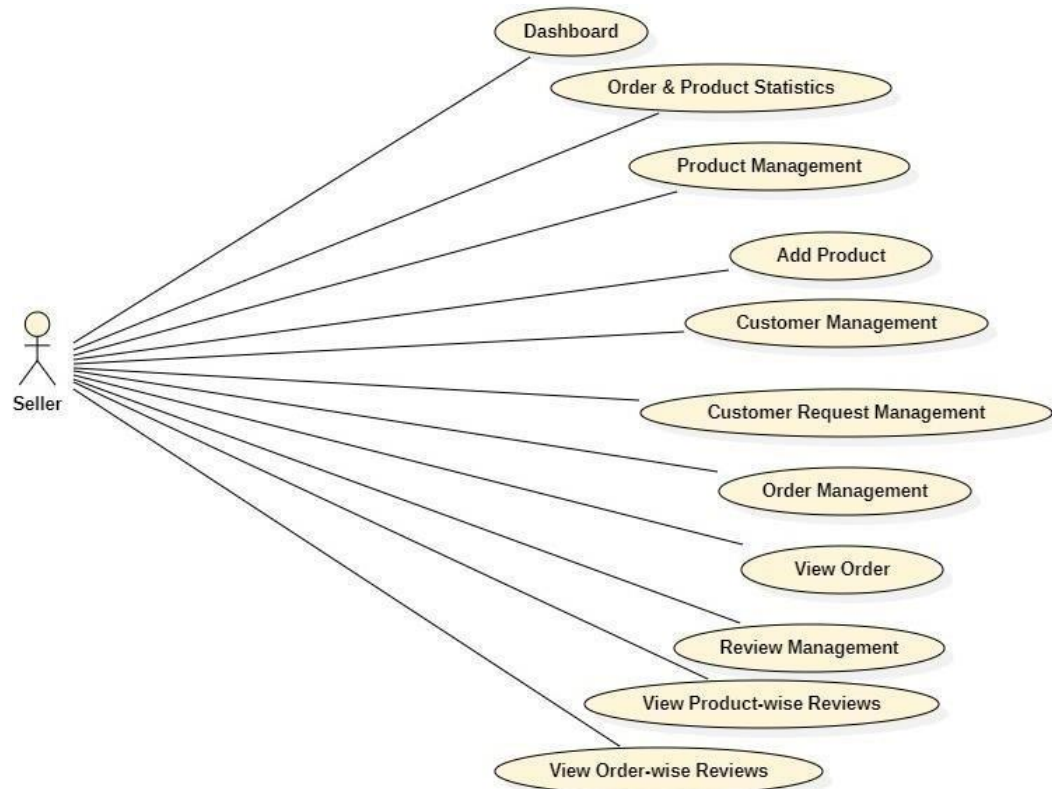
ADMIN



CUSTOMER



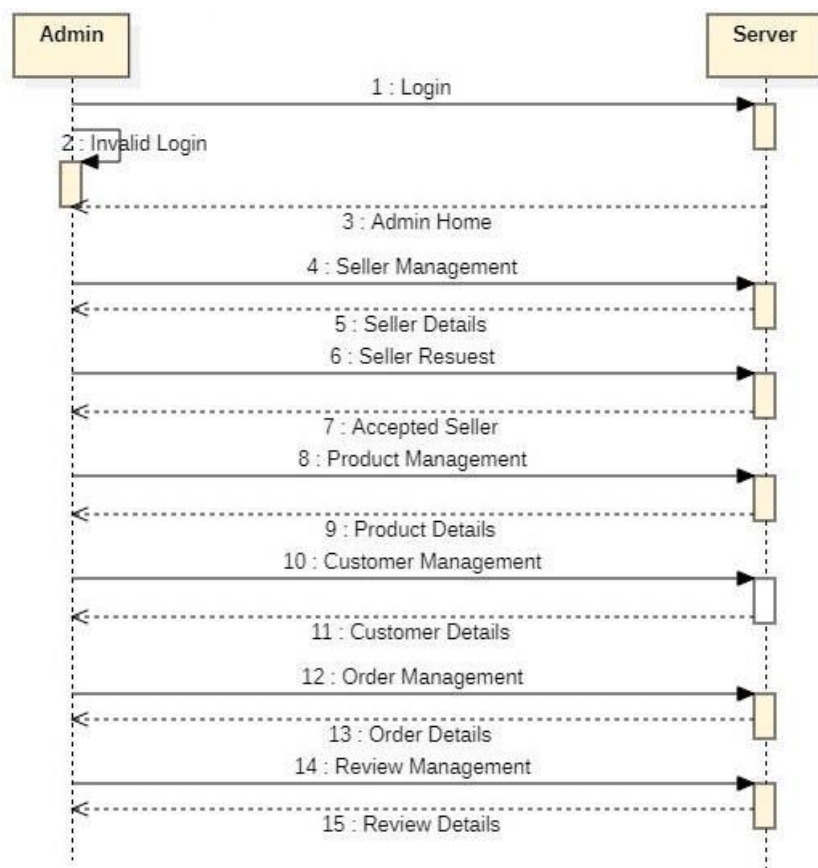
SELLER



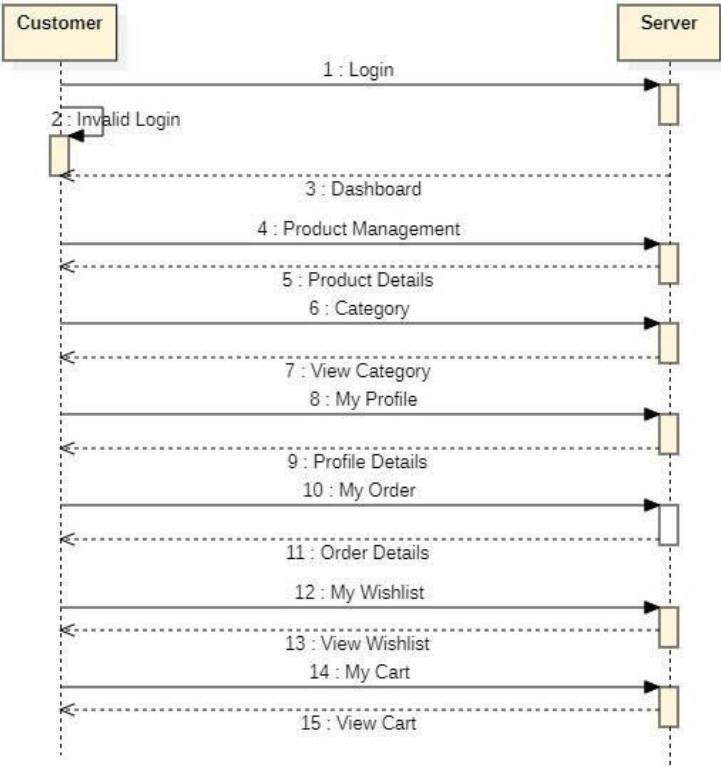
4.2.3 SEQUENCE DIAGRAM

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

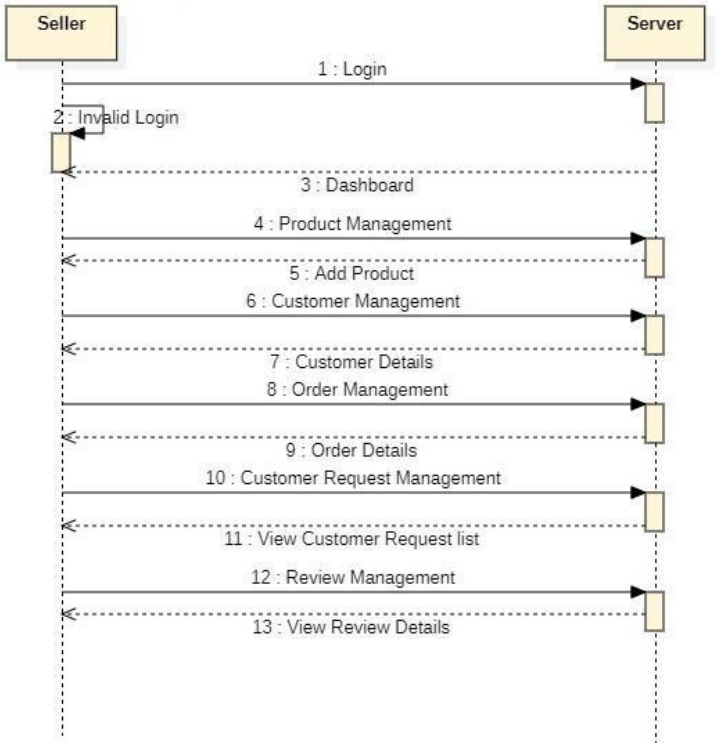
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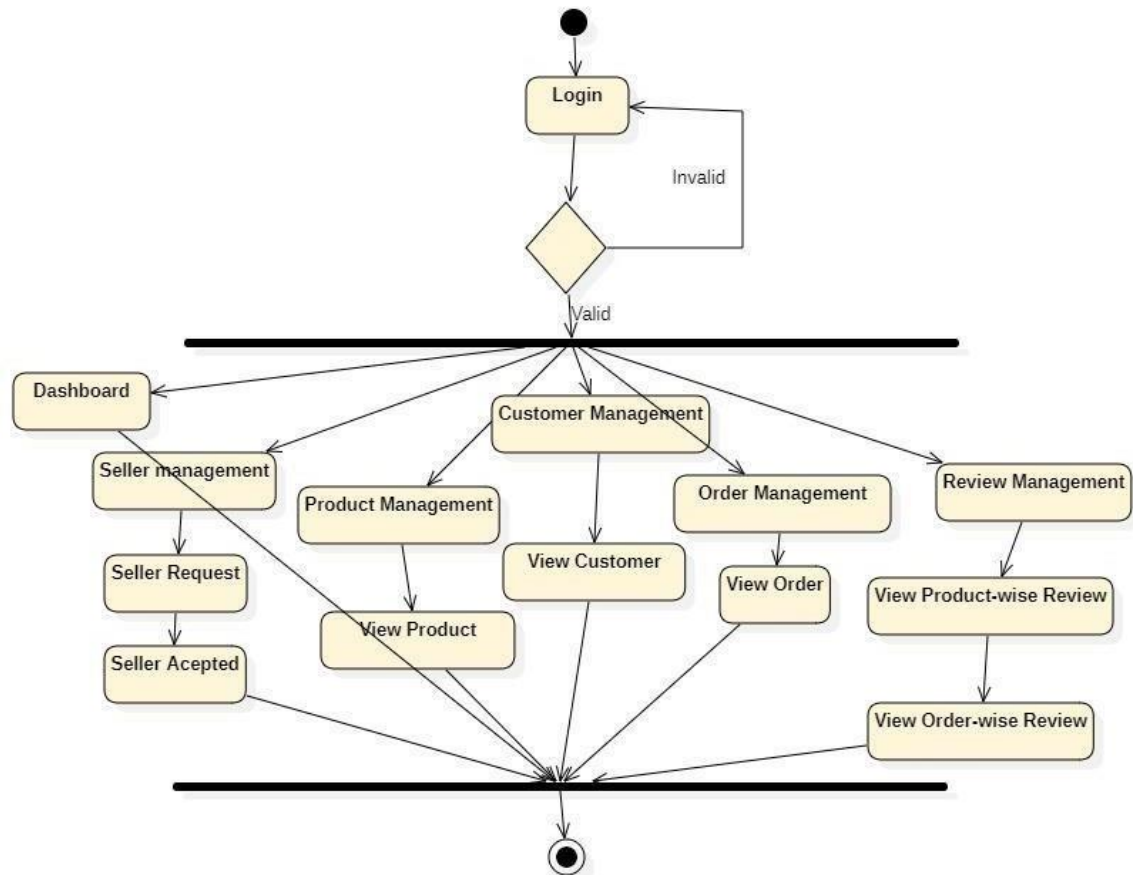
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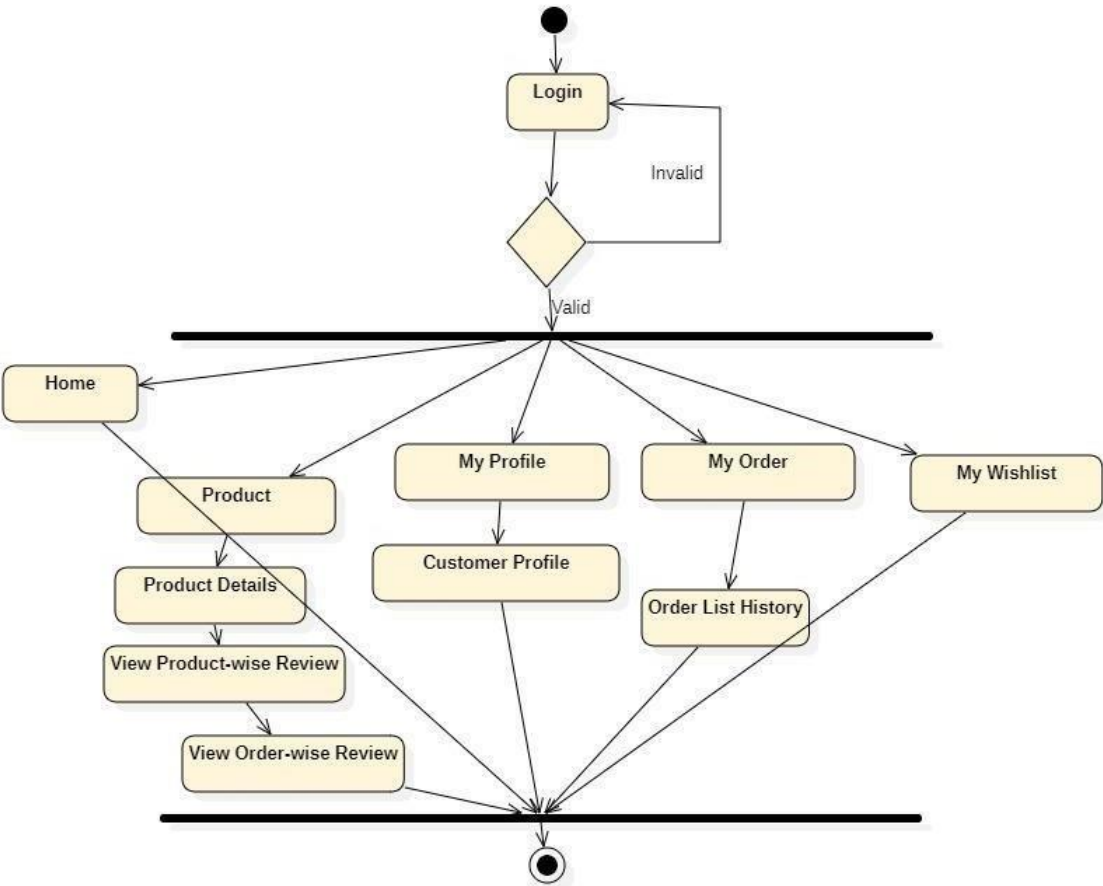
4.2.4 ACTIVITY DIAGRAM

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

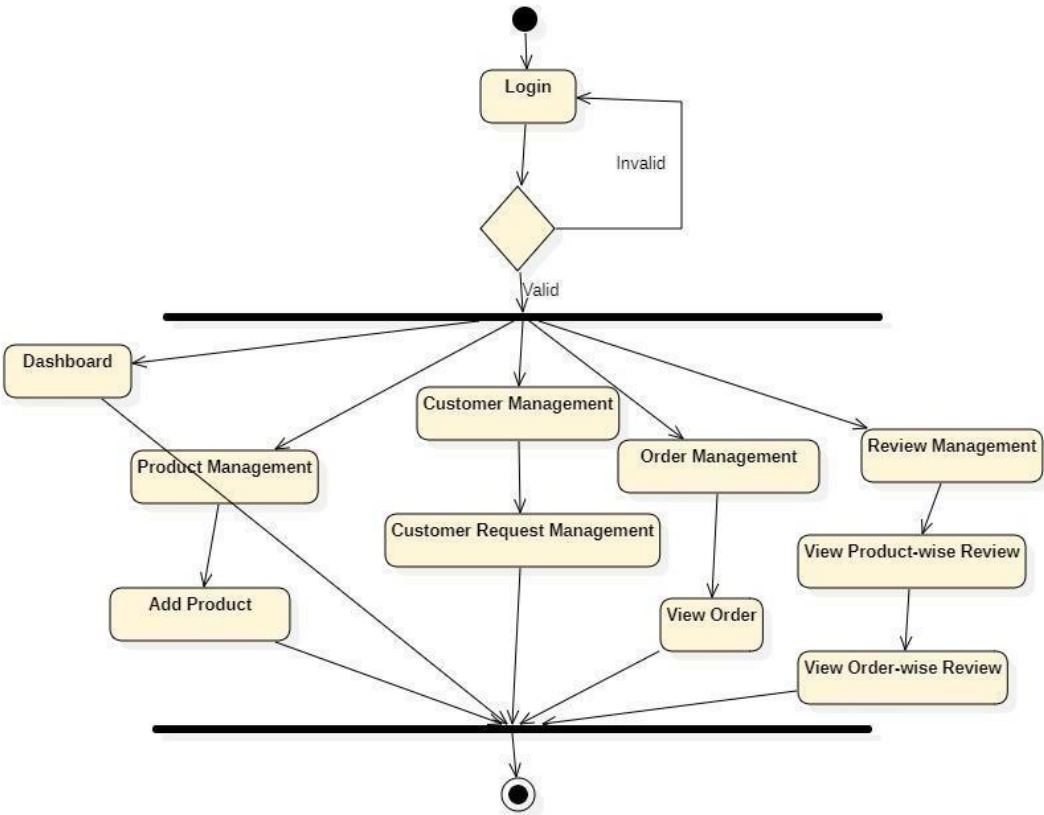
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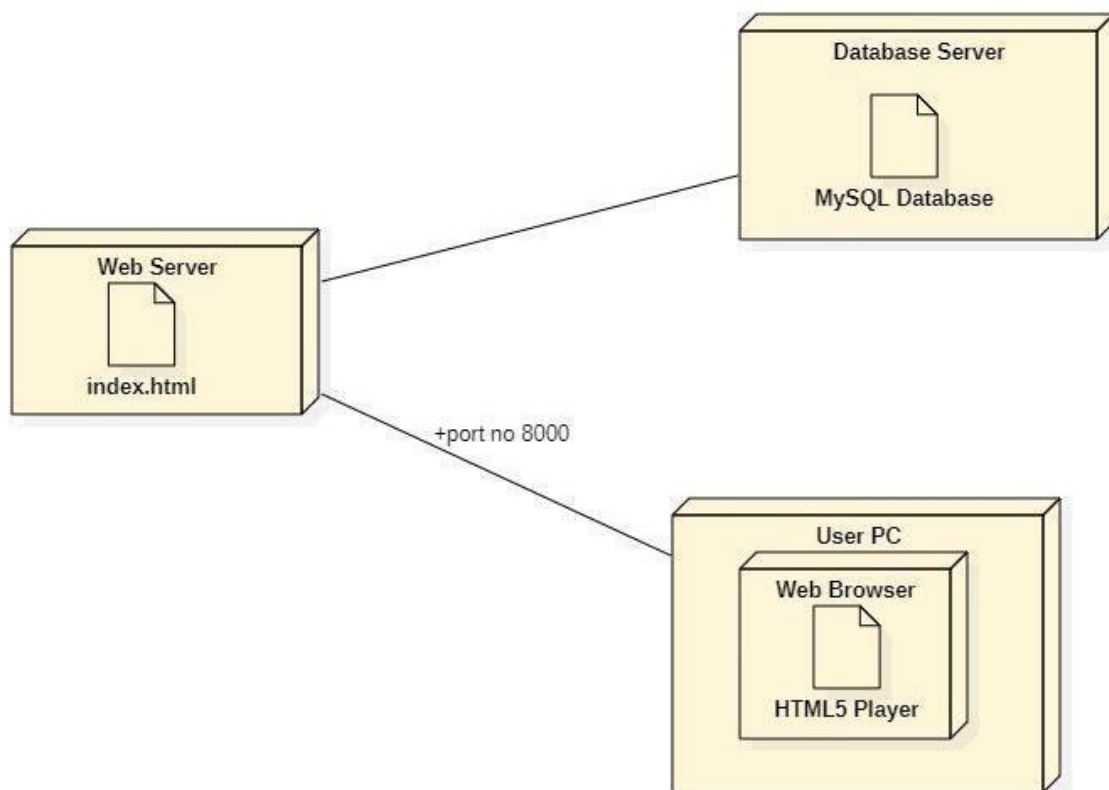


SELLER



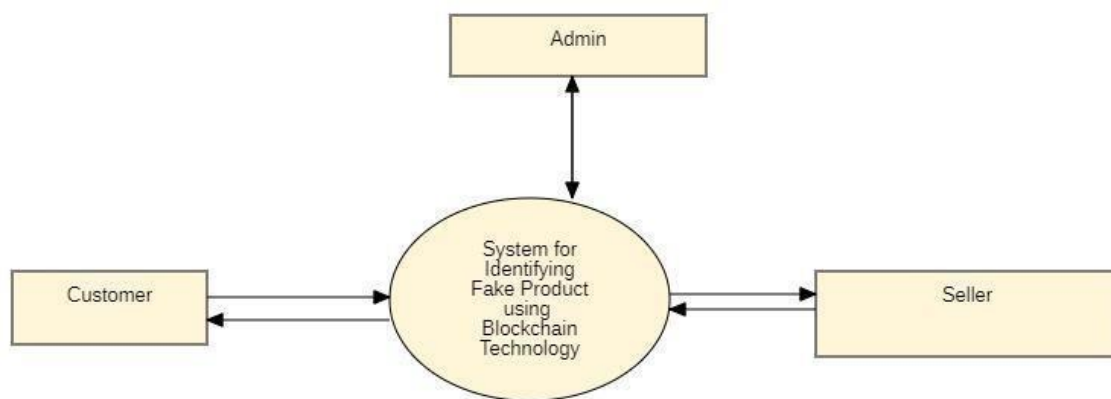
4.2.5 DEPLOYMENT DIAGRAM

Deployment diagram are a set of nodes and their relationships these nodes are physical entities where the components are displayed. Deployment diagram are used for visualizing the deployment view of a system. This is generally used by the deployment team. Deployment diagram show the relationship between the software and hardware components in the system and the physical distribution of the processing.



4.2.6 STATE CHART DIAGRAM

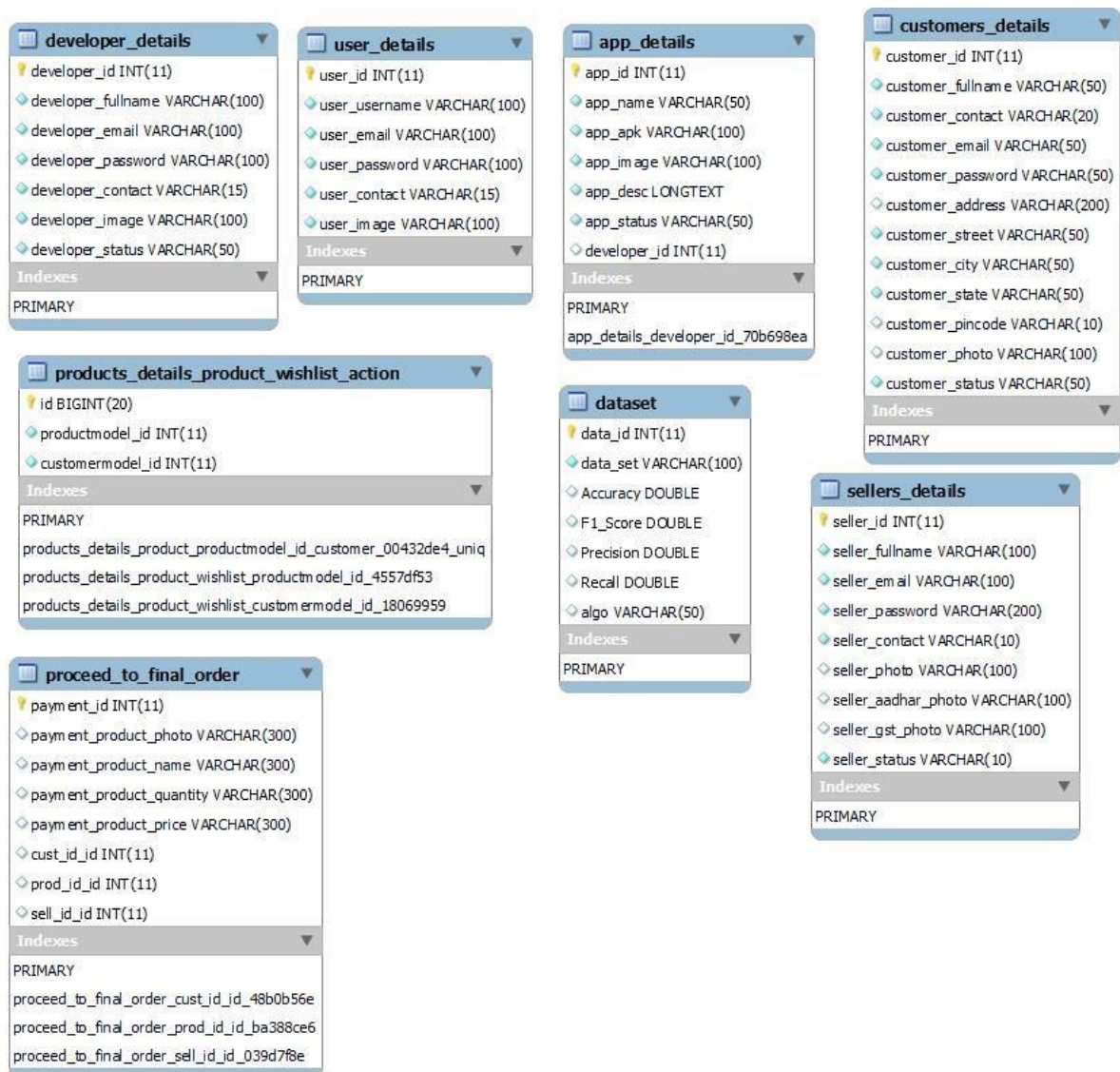
A state diagram is used to represent the condition of the system or part of the system at finite instances of time. It's a behavioral diagram and it represents the behavior using finite state transitions. State diagrams are also referred to as **State machines** and **State-chart Diagrams**. These terms are often used interchangeably. So simply, a state diagram is used to model the dynamic behavior of a class in response to time and changing external stimuli. We can say that each and every class has a state but we don't model every class using State diagrams. We prefer to model the states with three or more states.



4.2.7 ER DIAGRAM

The Entity Relational Model is a model for identifying entities to be represented in the database and representation of how those entities are related. The ER data model specifies enterprise schema that represents the overall logical structure of a database graphically.

The Entity Relationship Diagram explains the relationship among the entities present in the database. ER models are used to model real-world objects like a person, a car, or a company and the relation between these real-world objects. In short, the ER Diagram is the structural format of the database.



wishlist_details wishlist_id INT(11) product_wishlist_id INT(11) Indexes PRIMARY wishlist_details_product_wishlist_id_dcde411b	order_details order_id INT(11) order_product_price INT(11) order_product_quantity INT(11) order_product_amount INT(11) order_date DATETIME(6) order_status VARCHAR(300) order_address VARCHAR(300) order_customer_id INT(11) order_product_id INT(11) order_product_seller_id INT(11) order_seller_status VARCHAR(200) ofeedback_block VARCHAR(200) osentiment_block VARCHAR(200) ouser_block VARCHAR(200) qr_photo VARCHAR(100) Indexes PRIMARY order_details_order_customer_id_9993905f order_details_order_product_id_fd0274ff order_details_order_product_seller_id_5780ff45	products_feedbacks productwise_feedback_id INT(11) prodwise_expression VARCHAR(100) psentiment VARCHAR(100) custwise_comments_id INT(11) prodwise_comments_id INT(11) sellwise_comments_id INT(11) productwise_feedback VARCHAR(1000) productreview_date DATETIME(6) feedback_order_id INT(11) after_order_placed_revire_id INT(11) feedback_block VARCHAR(200) sentiment_block VARCHAR(200) user_block VARCHAR(200) Indexes PRIMARY feedback_details_custwise_comments_id_de0d5cfa feedback_details_prodwise_comments_id_bb6d2415 feedback_details_sellwise_comments_id_9932fa87 products_feedbacks_feedback_order_id_id_f5b569fc products_feedbacks_after_order_placed_revire_id_bf009d72	mycart_details cart_id INT(11) cart_product_quantity INT(11) cart_product_price INT(11) cart_owner_id INT(11) cart_product_id INT(11) cart_seller_id INT(11) Indexes PRIMARY mycart_details_cart_owner_id_6cc5bb15 mycart_details_cart_product_id_5f012321 mycart_details_cart_seller_id_25fae03b
products_details product_id INT(11) product_name VARCHAR(200) product_model VARCHAR(200) product_brand VARCHAR(200) product_category VARCHAR(200) product_original_price INT(11) product_discount_price INT(11) product_discount_percentage VARCHAR(50) product_photo1 VARCHAR(100) product_photo2 VARCHAR(100) product_photo3 VARCHAR(100) product_photo4 VARCHAR(100) product_photo5 VARCHAR(100) product_review LONGTEXT product_about VARCHAR(5000) product_creator_id INT(11) Indexes PRIMARY products_details_product_creator_id_47889052	feedback feedback_id INT(11) feedback LONGTEXT sentiment VARCHAR(50) rating INT(11) app_id INT(11) downloaded_user_id INT(11) Indexes PRIMARY feedback_app_id_2400651b feedback_downloaded_user_id_59e83f90	ordered_products_feedbacks order_review_id INT(11) orderproduct_review VARCHAR(1000) order_rating VARCHAR(100) order_expression VARCHAR(100) osentiment VARCHAR(100) customer_comments_id INT(11) order_product_comments_id INT(11) product_comments_id INT(11) seller_comments_id INT(11) Indexes PRIMARY customerapp_orderreviewmodels_customer_comments_id_2949f52a customerapp_orderreviewmodels_order_product_comments_id_ecd59da4 customerapp_orderreviewmodels_product_comments_id_cea621d6 customerapp_orderreviewmodels_seller_comments_id_fc0964f6	downloads download_id INT(11) app_id INT(11) downloaded_user_id INT(11) Indexes PRIMARY downloads_app_id_d85ee61a downloads_downloaded_user_id_b2e584ae
			url_deatils url_id INT(11) url LONGTEXT user_url_id INT(11) Indexes PRIMARY URL_Deatils_user_url_id_4397bdd4

4.2.8 DATA DICTIONARY DIAGRAM

auth_group

Table comments: auth_group

Column	Type	Null	Default
id	int (11)	No	
name	varchar (150)	No	

auth_group_permissions

Table comments: auth_group_permissions

Column	Type	Null	Default
id	bigint(20)	No	
group_id	int(11)	No	
permission_id	int(11)	No	

auth_permission

Table comments: auth_permission

Column	Type	Null	Default
id	int(11)	No	
name	varchar(255)	No	
content_type_id	int(11)	No	
codename	varchar(100)	No	

auth_user

Table comments: auth_user

Column	Type	Null	Default
id	int(11)	No	
password	varchar(128)	No	
last_login	datetime(6)	Yes	NULL
is_superuser	tinyint(1)	No	
username	varchar(150)	No	
first_name	varchar(150)	No	

last_name	varchar(150)	No	
email	varchar(254)	No	
is_staff	tinyint(1)	No	
is_active	tinyint(1)	No	
date_joined	datetime(6)	No	

auth_user_groups

Table comments: auth_user_groups

Column	Type	Null	Default
id	bigint(20)	No	
user_id	int(11)	No	
group_id	int(11)	No	

auth_user_user_permissions

Table comments: auth_user_user_permissions

Column	Type	Null	Default
id	bigint(20)	No	
user_id	int(11)	No	
permission_id	int(11)	No	

customers_details

Table comments: customers_details

Column	Type	Null	Default
customer_id	int(11)	No	
customer_fullname	varchar(50)	No	
customer_contact	varchar(20)	No	
customer_email	varchar(50)	No	
customer_password	varchar(50)	No	
customer_address	varchar(200)	Yes	NULL
customer_street	varchar(50)	No	
customer_city	varchar(50)	No	
customer_state	varchar(50)	No	

customer_pincode	varchar(10)	Yes	<i>NULL</i>
customer_photo	varchar(100)	Yes	<i>NULL</i>
customer_status	varchar(50)	No	

django_admin_log

Table comments: django_admin_log

Column	Type	Null	Default
id	int(11)	No	
action_time	datetime(6)	No	
object_id	longtext	Yes	<i>NULL</i>
object_repr	varchar(200)	No	
action_flag	smallint(5)	No	
change_message	longtext	No	
content_type_id	int(11)	Yes	<i>NULL</i>
user_id	int(11)	No	

django_content_type

Table comments: django_content_type

Column	Type	Null	Default
id	int(11)	No	
app_label	varchar(100)	No	
model	varchar(100)	No	

django_migrations

Table comments: django_migrations

Column	Type	Null	Default
Id	bigint(20)	No	
App	varchar(255)	No	
Name	varchar(255)	No	
Applied	datetime(6)	No	

mycart_details

Table comments: mycart_details

Column	Type	Null	Default
cart_id	int(11)	No	
cart_product_quantity	int(11)	Yes	<i>NULL</i>
cart_product_price	int(11)	Yes	<i>NULL</i>
cart_owner_id	int(11)	Yes	<i>NULL</i>
cart_product_id	int(11)	Yes	<i>NULL</i>
cart_seller_id	int(11)	Yes	<i>NULL</i>

order_details

Table comments: order_details

Column	Type	Null	Default
order_id	int(11)	No	
order_product_price	int(11)	Yes	<i>NULL</i>
order_product_quantity	int(11)	Yes	<i>NULL</i>
order_product_amount	int(11)	Yes	<i>NULL</i>
order_date	datetime(6)	No	
order_status	varchar(300)	No	
order_address	varchar(300)	Yes	<i>NULL</i>
order_customer_id	int(11)	No	
order_product_id	int(11)	Yes	<i>NULL</i>
order_product_seller_id	int(11)	Yes	<i>NULL</i>
order_seller_status	varchar(300)	No	
ofeedback_block	varchar(200)	Yes	<i>NULL</i>
osentiment_block	varchar(200)	Yes	<i>NULL</i>
ouser_block	varchar(200)	Yes	<i>NULL</i>
qr_photo	varchar(100)	Yes	<i>NULL</i>

ordered_products_feedbacks

Table comments: ordered_products_feedbacks

Column	Type	Null	Default
order_review_id	int(11)	No	

orderproduct_review	varchar(1000)	No	
order_rating	varchar(100)	Yes	<i>NULL</i>
order_expression	varchar(100)	Yes	<i>NULL</i>
Osentiment	varchar(100)	Yes	<i>NULL</i>
customer_comments_id	int(11)	Yes	<i>NULL</i>
order_product_comments_id	int(11)	Yes	<i>NULL</i>
product_comments_id	int(11)	Yes	<i>NULL</i>
seller_comments_id	int(11)	Yes	<i>NULL</i>

proceed_to_final_order

Table comments: proceed_to_final_order

Column	Type	Null	Default
payment_id	int(11)	No	
payment_product_photo	varchar(300)	Yes	<i>NULL</i>
payment_product_name	varchar(300)	Yes	<i>NULL</i>
payment_product_quantity	varchar(300)	Yes	<i>NULL</i>
payment_product_price	varchar(300)	Yes	<i>NULL</i>
cust_id_id	int(11)	Yes	<i>NULL</i>
prod_id_id	int(11)	Yes	<i>NULL</i>
sell_id_id	int(11)	Yes	<i>NULL</i>

products_details_product_wishlist_action

Table comments: products_details_product_wishlist_action

Column	Type	Null	Default
Id	bigint(20)	No	
productmodel_id	int(11)	No	
customermodel_id	int(11)	No	

products_feedbacks

Table comments: products_feedbacks

Column	Type	Null	Default
productwise_feedback_id	int(11)	No	
prodwise_expression	varchar(100)	No	

Psentiment	varchar(100)	Yes	<i>NULL</i>
custwise_comments_id	int(11)	Yes	<i>NULL</i>
prodwise_comments_id	int(11)	Yes	<i>NULL</i>
sellwise_comments_id	int(11)	Yes	<i>NULL</i>
productwise_feedback	varchar(1000)	Yes	<i>NULL</i>
productreview_date	datetime(6)	No	
feedback_order_id_id	int(11)	Yes	<i>NULL</i>
after_order_placed_revire_id	int(11)	Yes	<i>NULL</i>
feedback_block	varchar(200)	Yes	<i>NULL</i>
sentiment_block	varchar(200)	Yes	<i>NULL</i>
user_block	varchar(200)	Yes	<i>NULL</i>

sellers_details

Table comments: sellers_details

Column	Type	Null	Default
seller_id	int(11)	No	
seller_fullname	varchar(100)	No	
seller_email	varchar(100)	No	
seller_password	varchar(200)	No	
seller_contact	varchar(10)	No	
seller_photo	varchar(100)	Yes	<i>NULL</i>
seller_aadhar_photo	varchar(100)	Yes	<i>NULL</i>
seller_gst_photo	varchar(100)	Yes	<i>NULL</i>
seller_status	varchar(10)	No	

wishlist_details

Table comments: wishlist_details

Column	Type	Null	Default
wishlist_id	int(11)	No	
product_wishlist_id	int(11)	Yes	<i>NULL</i>

5. TECHNOLOGIES

5.1 WHAT IS PYTHON

Python is currently the most widely used multi-purpose, high-level programming language.

Python allows programming in Object-Oriented and Procedural paradigms. Python programs generally are smaller than other programming languages like Java.

Programmers have to type relatively less and indentation requirement of the language, makes them readable all the time.

Python language is being used by almost all tech-giant companies like – Google, Amazon, Facebook, Instagram, Dropbox, Uber... etc.

The biggest strength of Python is huge collection of standard library which can be used for the following –

- Blockchain
- GUI Applications (like Kivy, Tkinter, PyQt etc.)
- Web frameworks like Django (used by YouTube, Instagram, Dropbox)
- Image processing (like OpenCV, Pillow)
- Web scraping (like Scrapy, BeautifulSoup, Selenium)
- Test frameworks
- Multimedia

5.1.1 ADVANTAGES & DISADVANTAGES OF PYTHON

Advantages of Python:

1.Extensive Libraries

Python downloads with an extensive library and it contain code for various purposes like regular expressions, documentation-generation, unit-testing, web browsers, threading, databases, CGI, email, image manipulation, and more. So, we don't have to write the complete code for that manually.

2.Extensible

As we have seen earlier, Python can be extended to other languages. You can write some of your code in languages like C++ or C. This comes in handy, especially in projects.

3.Embeddable

Complimentary to extensibility, Python is embeddable as well. You can put your Python code in your source code of a different language, like C++. This lets us add scripting capabilities to our code in the other language.

4.IOT Opportunities

Since Python forms the basis of new platforms like Raspberry Pi, it finds the future bright for the Internet of Things. This is a way to connect the language with the real world.

5.Object-Oriented

This language supports both the procedural and object-oriented programming paradigms. While functions help us with code reusability, classes and objects let us model the real world.

ADVANTAGES OF PYTHON OVER OTHER LANGUAGES

1.Less Coding

Almost all of the tasks done in Python requires less coding when the same task is done in other languages. Python also has an awesome standard library support, so you don't have to search for any third-party libraries to get your job done.

2.Affordable

Python is free therefore individuals, small companies or big organizations can leverage the free available resources to build applications. Python is popular and widely used so it gives you better community support.

3.Python is for Everyone

Python code can run on any machine whether it is Linux, Mac or Windows. Programmers need to learn different languages for different jobs but with Python, you can professionally build web apps, perform data analysis and machine learning, automate things, do web scraping and also build games and powerful visualizations. It is an all-rounder programming language.

DISADVANTAGES OF PYTHON

1.Speed Limitations

We have seen that Python code is executed line by line. But since Python is interpreted, it often results in slow execution. This, however, isn't a problem unless speed is a focal point for the project. In other words, unless high speed is a requirement, the benefits offered by Python are enough to distract us from its speed limitations.

2.Weak in Mobile Computing and Browsers

While it serves as an excellent server-side language, Python is much rarely seen on the client-side. Besides that, it is rarely ever used to implement smartphone-based applications.

One such application is called Carbonnelle.

3.Design Restrictions

Python is dynamically-typed. This means that you don't need to declare the type of variable while writing the code. It uses duck-typing. But wait, what's that? Well, it just means that if it looks like a duck, it must be a duck. While this is easy on the programmers during coding, it can raise run-time errors.

4. Underdeveloped Database Access Layers

Compared to more widely used technologies like JDBC (Java Database Connectivity) and ODBC (Open Database Connectivity), Python's database access layers are a bit underdeveloped. Consequently, it is less often applied in huge enterprises.

5.Simple

Python's simplicity can indeed be a problem. Take my example. I don't do Java, I'm more of a Python person. To me, its syntax is so simple that the verbosity of Java code seems unnecessary.

5.1.2 HISTORY OF PYTHON

What do the alphabet and the programming language Python have in common? Right, both start with ABC. If we are talking about ABC in the Python context, it's clear that the programming language ABC is meant. ABC is a general-purpose programming language and programming environment, which had been developed in the Netherlands, Amsterdam, at the CWI (Centrum Wiskunde & Informatica). The greatest achievement of ABC was to influence the design of Python. Python was conceptualized in the late 1980s. Guido van

Rossum worked that time in a project at the CWI, called Amoeba, a distributed operating system. In an interview with Bill Venners¹, Guido van Rossum said: "In the early 1980s, I worked as an implementer on a team building a language called ABC at Centrum Wiskunde en Informatica (CWI). I don't know how well people know ABC's influence on Python. I try to mention ABC's influence because I'm indebted to everything I learned during that project and to the people who worked on it. Later on in the same Interview, Guido van Rossum continued: "I remembered all my experience and some of my frustration with ABC. I decided to try to design a simple scripting language that possessed some of ABC's better properties, but without its problems. So, I started typing. I created a simple virtual machine, a simple parser, and a simple runtime. I made my own version of the various ABC parts that I liked. I created a basic syntax, used indentation for statement grouping instead of curly braces or begin-end blocks, and developed a small number of powerful data types: a hash table (or dictionary, as we call it), a list, strings, and numbers."

5.2 BLOCKCHAIN TECHNOLOGY

Blockchain is a revolutionary decentralized digital ledger technology that fundamentally transforms the way data is stored, verified, and shared across a network of computers. It operates as a distributed ledger, where transactions are recorded in blocks that are linked together in a chronological and immutable chain. Each block contains a cryptographic hash of the previous block, ensuring the integrity of the data and making it nearly impossible to alter historical records without consensus from the network participants.

5.2.1 Categories:

Public Blockchain: This type of blockchain is open to anyone and allows for participation and transaction verification by the public. It offers a high level of transparency and decentralization, exemplified by cryptocurrencies like Bitcoin and Ethereum.

Private Blockchain: Access to a private blockchain is restricted to authorized users, often within a single organization or consortium of organizations. It provides more control over access and governance, making it suitable for enterprise applications such as supply chain management and financial services.

Consortium Blockchain: Controlled by a group of trusted entities, a consortium blockchain strikes a balance between the openness of public blockchains and the control of private ones.

It's commonly used in industries where multiple stakeholders collaborate on shared processes, such as healthcare and logistics.

5.2.2 Need for Blockchain:

Trust and Transparency: Blockchain technology eliminates the need for intermediaries by providing a transparent and immutable record of transactions. This fosters trust among participants and reduces the risk of fraud or manipulation.

Security: The cryptographic techniques used in blockchain ensure that data stored on the network is secure and tamper-proof. This enhances the overall security posture of applications built on blockchain technology.

Decentralization: By removing central points of control, blockchain decentralizes authority and distributes power among network participants. This resilience to single points of failure increases the reliability and availability of services.

Immutable Record-Keeping: Once data is recorded on the blockchain, it cannot be altered or deleted retroactively. This feature ensures the integrity and permanence of transaction history, providing a reliable audit trail.

5.2.3 Challenges:

Integration: Integrating blockchain technology with existing systems and processes can be challenging and may require significant changes to infrastructure and workflows.

Scalability: Some blockchain networks struggle to handle a large volume of transactions efficiently, leading to congestion and increased transaction costs. Scaling solutions, such as sharding and layer 2 protocols, are being developed to address this limitation.

Regulatory Uncertainty: The regulatory landscape surrounding blockchain and cryptocurrencies varies significantly from one jurisdiction to another. This regulatory uncertainty poses challenges for businesses and may hinder the widespread adoption of blockchain technology.

Privacy Concerns: While blockchain provides pseudonymity for users, ensuring data privacy without sacrificing transparency remains a challenge. Solutions such as zero-knowledge proofs and privacy-preserving techniques are being explored to address this issue.

5.2.4 Applications:

Product Authentication: Blockchain can be leveraged to verify the authenticity of products by recording their provenance and ownership history on an immutable ledger. This helps combat counterfeit goods and ensures consumer trust.

Supply Chain Management: Blockchain enables transparent and traceable supply chains by recording the movement of goods from their source to the end consumer. This enhances supply chain efficiency, reduces fraud, and improves accountability.

Digital Identity: Blockchain-based identity management systems offer a secure and verifiable way to manage digital identities, reducing the risk of identity theft and fraud. Individuals have greater control over their personal data, while businesses benefit from streamlined identity verification processes.

Track and Trace: Blockchain facilitates the tracking and tracing of goods throughout the supply chain, enabling consumers to verify the authenticity and origin of products.

5.2.5 Advantages:

Security: Blockchain's decentralized architecture and cryptographic techniques ensure the integrity and security of data stored on the network. Transactions are cryptographically secured and tamper-proof, reducing the risk of fraud and unauthorized access.

Transparency: Blockchain provides a transparent and auditable record of transactions, visible to all network participants. This transparency fosters trust among users and enhances accountability.

Decentralization: Blockchain eliminates the need for intermediaries and central authorities, reducing dependency on trusted third parties and enabling peer-to-peer transactions. This decentralization enhances resilience and ensures continuity of services.

Trust: Blockchain's immutable ledger and consensus mechanisms create a trustless environment where participants can transact with confidence, knowing that the integrity of the system is maintained by cryptographic principles.

DISADVANTAGES:

Scalability Issues: Some blockchain networks struggle to handle a high volume of transactions, leading to congestion, increased transaction times, and higher fees. Scalability solutions, such as off-chain scaling and layer 2 protocols, are being developed to address this limitation.

Regulatory Challenges: The regulatory landscape surrounding blockchain and cryptocurrencies is still evolving and varies significantly from one jurisdiction to another. Regulatory uncertainty poses challenges for businesses, stifles innovation, and may deter investment in blockchain projects.

Privacy Concerns: While blockchain provides pseudonymity for users, ensuring data privacy without sacrificing transparency is a significant challenge. Solutions such as zero-knowledge proofs and privacy-preserving techniques are being explored to address privacy concerns and enhance confidentiality on the blockchain.

Lack of Interoperability: Interoperability between different blockchain platforms and protocols remains a challenge, hindering the seamless exchange of data and assets across disparate networks. Standards and protocols for interoperability are being developed to facilitate communication and interoperability between blockchain ecosystems.

5.3 MODULES USED IN PYTHON

Django:

Django serves as the framework for building the backend of your system. It provides tools for handling web requests, managing data models, and rendering HTML templates. In the context of a product authentication system, Django can handle user authentication, product registration, and verification processes.

User Authentication: Implement user authentication to allow different types of users, such as manufacturers, distributors, and consumers, to access the system securely.

Product Registration: Create forms and views to allow manufacturers to register genuine products onto the blockchain. Django's form handling capabilities can be used to validate input data before recording it on the blockchain.

Product Verification: Develop endpoints or views for users to verify the authenticity of products by querying the blockchain. Django can handle the logic for retrieving product information and displaying verification results to users.

NumPy:

NumPy is used for numerical computations and data manipulation. While blockchain technology handles the secure storage and verification of product data, NumPy can be employed for analyzing and processing data related to product attributes and supply chain information.

Data Preprocessing: Utilize NumPy to preprocess and clean supply chain data before analysis. This may involve tasks such as handling missing values, normalizing data, or encoding categorical variables.

Statistical Analysis: Perform statistical analysis on supply chain data to identify patterns or anomalies indicative of counterfeit products. NumPy's functions for calculating descriptive statistics, correlations, and distributions can be helpful in this regard.

Numerical Computations: Use NumPy for numerical computations required in product authentication algorithms or fraud detection models. This could involve mathematical operations on arrays representing product attributes or transaction data.

Pandas:

Pandas is a powerful library for data manipulation and analysis, particularly suited for working with structured data. In the context of a product authentication system, Pandas can be used for managing and analyzing large datasets related to product registrations, supply chain transactions, and verification logs.

Data Handling: Load and manipulate supply chain data stored in various formats (e.g., CSV, Excel) using Pandas Data Frames. Pandas provides convenient functions for reading, writing, and querying data from different sources.

Exploratory Data Analysis (EDA): Conduct exploratory data analysis on supply chain data to gain insights into product distribution patterns, transaction frequencies, or other relevant metrics. Pandas' capabilities for data manipulation, aggregation, and visualization support this process.

5.4 INSTALL PYTHON ON WINDOWS

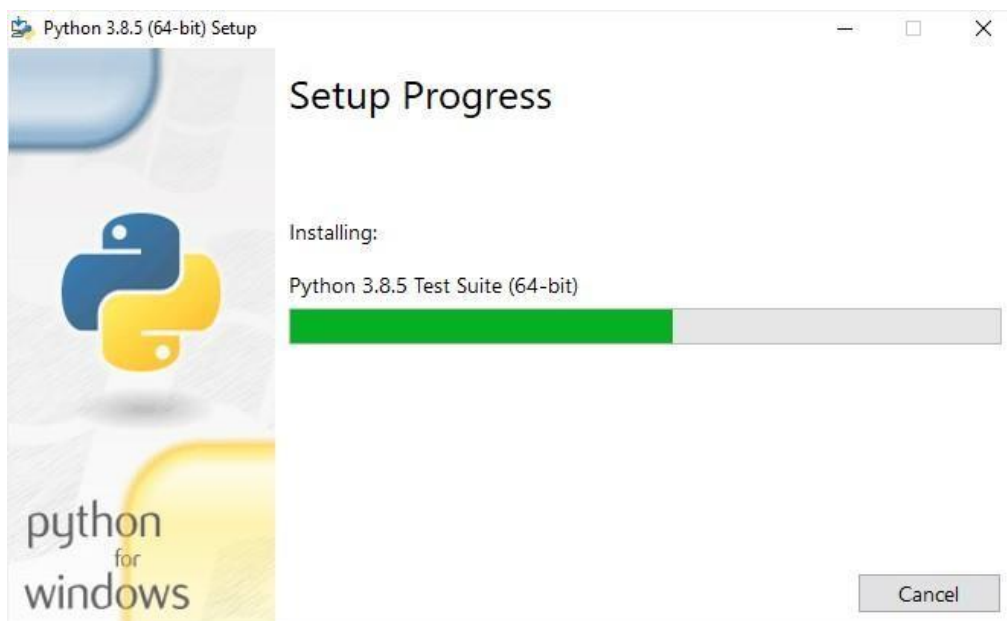
First, download the latest version of Python from the download page.

Second, double-click the installer file to launch the setup wizard.

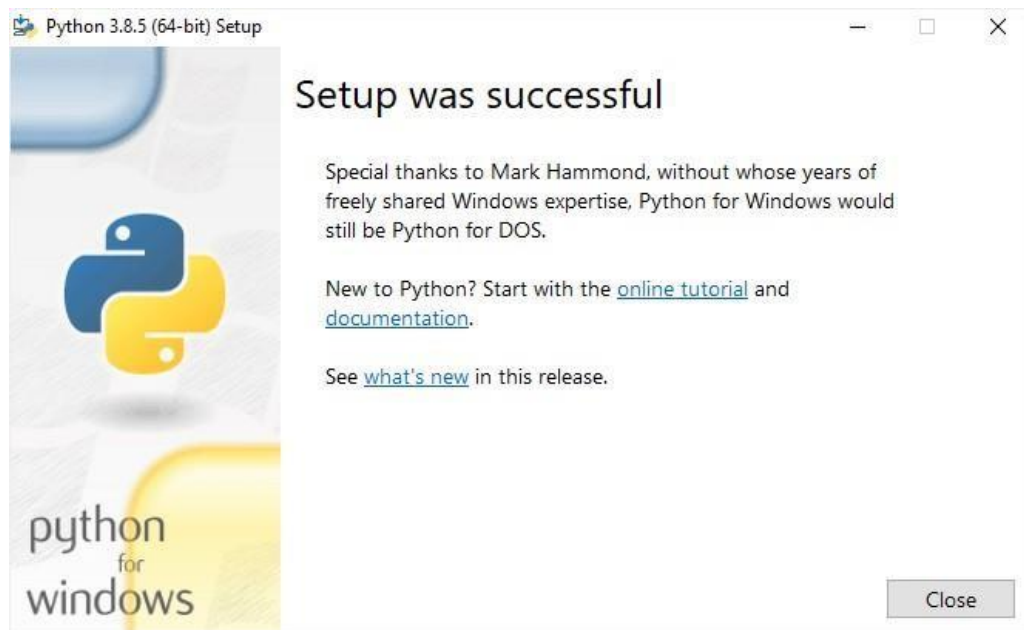
In the setup window, you need to check the **Add Python 3.8 to PATH** and click Install Now to begin the installation.



It'll take a few minutes to complete the setup.

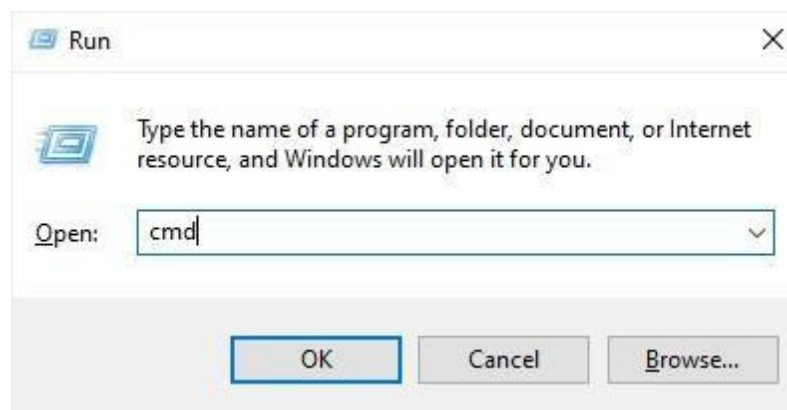


Once the setup completes, you'll see the following window:

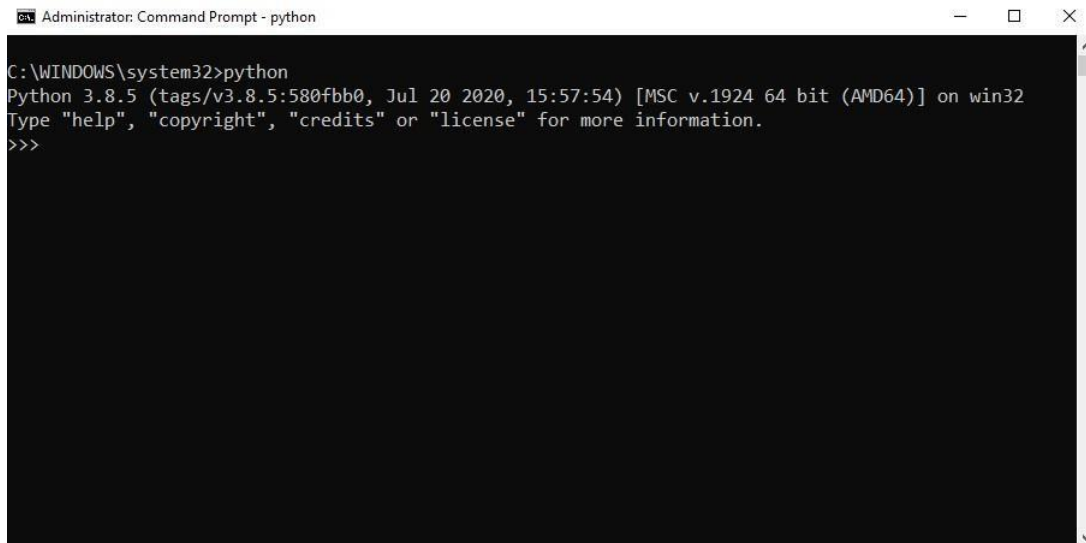


Verify the installation

To verify the installation, you open the Run window and type cmd and press Enter:



In the Command Prompt, type python command as follows:



```
C:\WINDOWS\system32>python
Python 3.8.5 (tags/v3.8.5:580fbb0, Jul 20 2020, 15:57:54) [MSC v.1924 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

If you see the output like the above screenshot, you've successfully installed Python on your computer.

To exit the program, you type Ctrl-Z and press Enter.

If you see the following output from the Command Prompt after typing the python command:

```
'python' is not recognized as an internal or external command,
operable
program or batch file.
```

Likely, you didn't check the **Add Python 3.8 to PATH** checkbox when you install Python.

Install Python on macOS

It's recommended to install Python on macOS using an official installer. Here are the steps:

- First, download a Python release for macOS.
- Second, run the installer by double-clicking the installer file.
- Third, follow the instruction on the screen and click the Next button until the installer completes.

Install Python on Linux

Before installing Python 3 on your Linux distribution, you check whether Python 3 was already installed by running the following command from the terminal:

```
python3 --version
```

If you see a response with the version of Python, then your computer already has Python 3 installed. Otherwise, you can install Python 3 using a package management system.

For example, you can install Python 3.10 on Ubuntu using apt:

```
sudo apt install python3.10
```

To install the newer version, you replace 3.10 with that version.

A quick introduction to the Visual Studio Code

Visual Studio Code is a lightweight source code editor. The Visual Studio Code is often called VS Code. The VS Code runs on your desktop. It's available for Windows, macOS, and Linux. VS Code comes with many features such as IntelliSense, code editing, and extensions that allow you to edit Python source code effectively. The best part is that the VS Code is opensource and free.

Besides the desktop version, VS Code also has a browser version that you can use directly in your web browser without installing it.

This tutorial teaches you how to set up Visual Studio Code for a Python environment so that you can edit, run, and debug Python code.

Setting up Visual Studio Code

To set up the VS Code, you follow these steps:

First, navigate to the VS Code official website and download the VS code based on your platform (Windows, macOS, or Linux).

Second, launch the setup wizard and follow the steps.

Once the installation completes, you can launch the VS code application:

6. IMPLEMENTATION

6.1 SOFTWARE ENVIRONMENT

6.1.1 PYTHON

Python is a general-purpose interpreted, interactive, object-oriented, and high-level programming language. An interpreted language, Python has a design philosophy that emphasizes code readability (notably using whitespace indentation to delimit code blocks rather than curly brackets or keywords), and a syntax that allows programmers to express concepts in fewer lines of code than might be used in languages such as C++ or Java. It provides constructs that enable clear programming on both small and large scales. Python interpreters are available for many operating systems. C, Python, the reference implementation of Python, is open-source software and has a community-based development model, as do nearly all of its variant implementations. C, Python is managed by the non-profit Python Software Foundation. Python features a dynamic type system and automatic memory management. Interactive Mode Programming.

6.1.2 SAMPLE CODE

BlockchainAlgo.py import hashlib class

```
HashDataBlock: def __init__(self,
previous_block_hash, data_list):
self.previous_block_hash = previous_block_hash
    self.data_list = data_list
    self.block_data = "_".join(data_list) + "_" + previous_block_hash
self.block_hash = hashlib.sha256(self.block_data.encode()).hexdigest()
```

settings.py

```
from pathlib import Path import os
BASE_DIR = Path(__file__).resolve().parent.parent
```

```

SECRET_KEY = 'django-insecure-
x8n5_eoa+yh0h$_)=l7n^c2if#n6c% % % %m8g!)b5y4m4c8sq' DEBUG = True
ALLOWED_HOSTS = []
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'adminapp',
    'customerapp',
    'sellerapp',
    'mainapp'
]
MIDDLEWARE = [
    'django.middleware.security.SecurityMiddleware',
    'django.contrib.sessions.middleware.SessionMiddleware',
    'django.middleware.common.CommonMiddleware',
    'django.middleware.csrf.CsrfViewMiddleware',
    'django.contrib.auth.middleware.AuthenticationMiddleware',
    'django.contrib.messages.middleware.MessageMiddleware',
    'django.middleware.clickjacking.XFrameOptionsMiddleware',
]
ROOT_URLCONF = 'fake_product_detection.urls'
TEMPLATES = [
    {
        'BACKEND': 'django.template.backends.django.DjangoTemplates',
        'DIRS': [os.path.join(BASE_DIR, 'assets/templates')],
        'APP_DIRS': True,
        'OPTIONS': {
            'context_processors': [
                'django.template.context_processors.debug',
                'django.template.context_processors.request',

```

```

        'django.contrib.auth.context_processors.auth',
        'django.contrib.messages.context_processors.messages',
    ],
},
},
]
WSGI_APPLICATION = 'fake_product_detection.wsgi.application'
DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.mysql',
        'NAME': 'ecommerce_fake_reviews',
        'HOST': '127.0.0.1',
        'USER': 'root',
        'PASSWORD': '3000',
        'PORT': '3306',
        'OPTION': {
            'init_command': "SET sql-mode = 'STRICT-TRANS_TABLES'"
        }
    }
}
AUTH_PASSWORD_VALIDATORS = [
    {
        'NAME': 'django.contrib.auth.password_validation.UserAttributeSimilarityValidator',
    },
    {
        'NAME': 'django.contrib.auth.password_validation.MinimumLengthValidator',
    },
    {
        'NAME': 'django.contrib.auth.password_validation.CommonPasswordValidator',
    },
    {
        'NAME': 'django.contrib.auth.password_validation.NumericPasswordValidator',
    },
]

```

```

LANGUAGE_CODE = 'en-us'
TIME_ZONE = 'UTC'
USE_I18N = True
USE_TZ = True
STATIC_URL = 'static/'
STATICFILES_DIRS = os.path.join(BASE_DIR, 'assets/static'),
MEDIA_URL = '/media/'
MEDIA_ROOT = os.path.join(BASE_DIR, 'media')
DEFAULT_AUTO_FIELD = 'django.db.models.BigAutoField'

```

Views.py

```

from django.shortcuts import
render, redirect, get_object_or_404 from django.contrib import
messages from customerapp.models import * from
customerapp.BlockchainAlgo import HashDataBlock from
sellerapp.models import * # from adminapp.models import * #
Create your views here. def main_admin_login(request):    if
request.method == "POST":
    username=request.POST.get("username")
password=request.POST.get("password")    if
username == "admin" and password == "admin":
messages.success(request, 'Successfully Login')
return redirect('admin-home')
    else:
        messages.warning(request, 'invalid login')
return redirect('main-admin-login')    return
render(request, "main/main-admin-login.html")    def
admin_home(request):
    order1=OrderModels.objects.count()
seller=SellerModel.objects.filter(seller_status="pending").count()
seller1=SellerModel.objects.filter(seller_status="accepted").count()
seller2=SellerModel.objects.filter(seller_status="declined").count()

```

```

fakereviews=FeedbackModel.objects.count()
genuinereviews=OrderReviewModels.objects.count()    return
render(request,"admin/admin-home.html", def
admin_sellers_request(request):
    s1=SellerModel.objects.order_by("-seller_id")    return
render(request,"admin/admin-sellers-request.html",{ 's':s1 }) def
accept_seller(request,id):
    accept = get_object_or_404(SellerModel,seller_id=id)
accept.seller_status = "accepted"
accept.save(update_fields=["seller_status"])
accept.save()    return redirect('admin-sellers-request')
def decline_seller(request,id):
    decline = get_object_or_404(SellerModel,seller_id=id)
decline.seller_status = "declined"
decline.save(update_fields=["seller_status"])    decline.save()
    return redirect('admin-sellers-request') def
admin_all_sellers(request):
    return render(request,"admin/admin-all-sellers.html",{ 's':s1 }) def
admin_edit_sellers(request):
    return render(request,"admin/admin-edit-sellers.html") def
admin_delete_sellers(request):
    return render(request,"admin/admin-delete-sellers.html")
def admin_view_product(request):
    s1 = ProductModel.objects.all().order_by('-product_id')    return
render(request,"admin/admin-view-product.html",{ 's':s1 }) def
admin_view_customers(request):    c1 = CustomerModel.objects.all()
return render(request,"admin/admin-view-customers.html",{ 'c':c1 })
def admin_view_order(request):    a=OrderModels.objects.all()
return render(request,"admin/admin-view-order.html",{ 'a':a }) def
admin_view_order_details(request):
    return render(request,"admin/admin-view-order-details.html") def
admin_view_feedback_orderwise(request):
}

```


7. SYSTEM TESTING

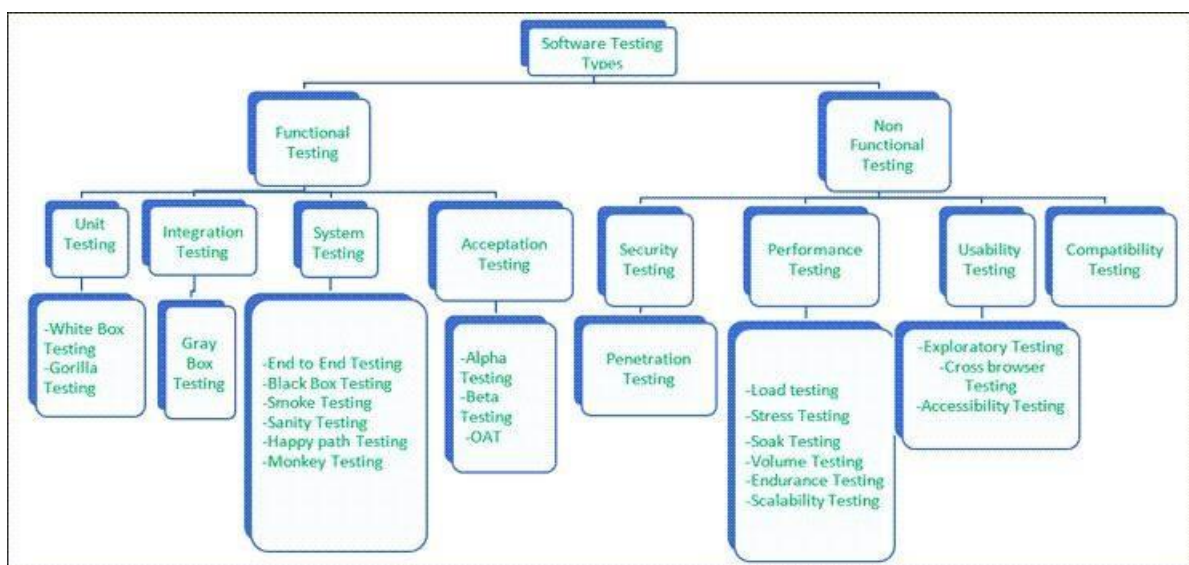
7.1 INTRODUCTION TO TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

7.2 TYPES OF SOFTWARE TESTING

We, as testers, are aware of the various types of Software Testing like Functional Testing, Non-Functional Testing, Automation Testing, Agile Testing, and their sub-types, etc. Each type of testing has its own features, advantages, and disadvantages as well. However, in this tutorial, we have covered mostly each and every type of software testing which we usually use in our day-to-day testing life.

Different Types of Software Testing



Functional Testing

There are four main types of functional testing.

1) Unit Testing

Unit testing is a type of software testing which is done on an individual unit or component to test its corrections. Typically, Unit testing is done by the developer at the application development phase. Each unit in unit testing can be viewed as a method, function, procedure, or object. Developers often use test automation tools such as NUnit, Xunit, JUnit for the test execution.

Unit testing is important because we can find more defects at the unit test level.

For example, there is a simple calculator application. The developer can write the unit test to check if the user can enter two numbers and get the correct sum for addition functionality.

a) White Box Testing

White box testing is a test technique in which the internal structure or code of an application is visible and accessible to the tester. In this technique, it is easy to find loopholes in the design of an application or fault in business logic. Statement coverage and decision coverage/branch coverage are examples of white box test techniques.

b) Gorilla Testing

Gorilla testing is a test technique in which the tester and/or developer test the module of the application thoroughly in all aspects. Gorilla testing is done to check how robust your application is.

For example, the tester is testing the pet insurance company's website, which provides the service of buying an insurance policy, tag for the pet, Lifetime membership. The tester can focus on any one module, let's say, the insurance policy module, and test it thoroughly with positive and negative test scenarios.

2) Integration Testing

Integration testing is a type of software testing where two or more modules of an application are logically grouped together and tested as a whole. The focus of this type of testing

is to find the defect on interface, communication, and data flow among modules. Topdown or Bottom-up approach is used while integrating modules into the whole system.

This type of testing is done on integrating modules of a system or between systems. **For example**, a user is buying a flight ticket from any airline website. Users can see flight details and payment information while buying a ticket, but flight details and payment processing are two different systems. Integration testing should be done while integrating of airline website and payment processing system.

a) Gray box testing

As the name suggests, gray box testing is a combination of white-box testing and blackbox testing. Testers have partial knowledge of the internal structure or code of an application.

3) System Testing

System testing is types of testing where tester evaluates the whole system against the specified requirements.

a) End to End Testing

It involves testing a complete application environment in a situation that mimics realworld use, such as interacting with a database, using network communications, or interacting with other hardware, applications, or systems if appropriate.

For example, a tester is testing a pet insurance website. End to End testing involves testing of buying an insurance policy, LPM, tag, adding another pet, updating credit card information on users' accounts, updating user address information, receiving order confirmation emails and policy documents.

b) Black Box Testing

Blackbox testing is a software testing technique in which testing is performed without knowing the internal structure, design, or code of a system under test. Testers should focus only on the input and output of test objects.

Detailed information about the advantages, disadvantages, and types of Black Box testing can be found [here](#).

c) Smoke Testing

Smoke testing is performed to verify that basic and critical functionality of the system under test is working fine at a very high level.

Whenever a new build is provided by the development team, then the Software Testing team validates the build and ensures that no major issue exists. The testing team will ensure that the build is stable, and a detailed level of testing will be carried out further.

For example, tester is testing pet insurance website. Buying an insurance policy, adding another pet, providing quotes are all basic and critical functionality of the application. Smoke testing for this website verifies that all these functionalities are working fine before doing any in-depth testing.

d) Sanity Testing

Sanity testing is performed on a system to verify that newly added functionality or bug fixes are working fine. Sanity testing is done on stable build. It is a subset of the regression test.

For example, a tester is testing a pet insurance website. There is a change in the discount for buying a policy for second pet. Then sanity testing is only performed on buying insurance policy module.

e) Happy path Testing

The objective of Happy Path Testing is to test an application successfully on a positive flow. It does not look for negative or error conditions. The focus is only on valid and positive inputs through which the application generates the expected output.

f) Monkey Testing

Monkey Testing is carried out by a tester, assuming that if the monkey uses the application, then how random input and values will be entered by the Monkey without any knowledge or understanding of the application.

The objective of Monkey Testing is to check if an application or system gets crashed by providing random input values/data. Monkey Testing is performed randomly, no test cases are scripted, and it is not necessary to be aware of the full functionality of the system.

4) Acceptance Testing

Acceptance testing is a type of testing where client/business/customer test the software with real time business scenarios.

The client accepts the software only when all the features and functionalities work as expected. This is the last phase of testing, after which the software goes into production. This is also called User Acceptance Testing (UAT).

a) Alpha Testing

Alpha testing is a type of acceptance testing performed by the team in an organization to find as many defects as possible before releasing software to customers.

For example, the pet insurance website is under UAT. UAT team will run real-time scenarios like buying an insurance policy, buying annual membership, changing the address, ownership transfer of the pet in a same way the user uses the real website. The team can use test credit card information to process payment-related scenarios.

b) Beta Testing

Beta Testing is a type of software testing which is carried out by the clients/customers. It is performed in the **Real Environment** before releasing the product to the market for the actual end-users.

Beta Testing is carried out to ensure that there are no major failures in the software or product, and it satisfies the business requirements from an end-user perspective. Beta Testing is successful when the customer accepts the software.

Usually, this testing is typically done by the end-users. This is the final testing done before releasing the application for commercial purposes. Usually, the Beta version of the software or product released is limited to a certain number of users in a specific area.

So, the end-user uses the software and shares the feedback with the company. The company then takes necessary action before releasing the software worldwide.

c) Operational acceptance testing (OAT)

Operational acceptance testing of the system is performed by operations or system administration staff in the production environment. The purpose of operational acceptance testing is to make sure that the system administrators can keep the system working properly for the users in a real-time environment.

The focus of the OAT is on the following points:

- Testing of backup and restore.
- Installing, uninstalling, upgrading software.
- The recovery process in case of natural disaster.
- User management.
- Maintenance of the software. **Non-Functional Testing**

There are four main types of functional testing.

1) Security Testing

It is a type of testing performed by a special team. Any hacking method can penetrate the system.

Security Testing is done to check how the software, application, or website is secure from internal and/or external threats. This testing includes how much software is secure from malicious programs, viruses and how secure & strong the authorization and authentication processes are.

It also checks how software behaves for any hacker's attack & malicious programs and how software is maintained for data security after such a hacker attack.

a) Penetration Testing

Penetration Testing or Pen testing is the type of security testing performed as an authorized cyberattack on the system to find out the weak points of the system in terms of security.

Pen testing is performed by outside contractors, generally known as ethical hackers. That is why it is also known as ethical hacking. Contractors perform different operations like SQL injection, URL manipulation, Privilege Elevation, session expiry, and provide reports to the organization.

Notes: Do not perform the Pen testing on your laptop/computer. Always take written permission to do pen tests.

2) Performance Testing

Performance testing is testing of an application's stability and response time by applying load.

The word stability means the ability of the application to withstand in the presence of load. Response time is how quickly an application is available to users. Performance testing is done with the help of tools. Loader.IO, JMeter, LoadRunner, etc. are good tools available in the market.

a) Load testing

Load testing is testing of an application's stability and response time by applying load, which is equal to or less than the designed number of users for an application.

For example, your application handles 100 users at a time with a response time of 3 seconds, then load testing can be done by applying a load of the maximum of 100 or less than 100 users. The goal is to verify that the application is responding within 3 seconds for all the users. **b)**

Stress Testing

Stress testing is testing an application's stability and response time by applying load, which is more than the designed number of users for an application.

For example, your application handles 1000 users at a time with a response time of 4 seconds, then stress testing can be done by applying a load of more than 1000 users. Test the application with 1100,1200,1300 users and notice the response time. The goal is to verify the stability of an application under stress.

c) Scalability Testing

Scalability testing is testing an application's stability and response time by applying load, which is more than the designed number of users for an application.

For example, your application handles 1000 users at a time with a response time of 2 seconds, then scalability testing can be done by applying a load of more than 1000 users and gradually increasing the number of users to find out where exactly my application is crashing.

Let's say my application is giving response time as follows:

- 1000 users -2 sec
- 1400 users -2 sec
- 4000 users -3 sec
- 5000 users -45 sec
- 5150 users- crash – This is the point that needs to identify in scalability testing

d) Volume testing (flood testing)

Volume testing is testing an application's stability and response time by transferring a large volume of data to the database. Basically, it tests the capacity of the database to handle the data.

e) Endurance Testing (Soak Testing)

Endurance testing is testing an application's stability and response time by applying load continuously for a longer period to verify that the application is working fine.

For example, car companies soak testing to verify that users can drive cars continuously for hours without any problem.

3) Usability Testing

Usability testing is testing an application from the user's perspective to check the look and feel and user-friendliness.

For example, there is a mobile app for stock trading, and a tester is performing usability testing. Testers can check the scenario like if the mobile app is easy to operate with one hand

or not, scroll bar should be vertical, background color of the app should be black and price of and stock is displayed in red or green colour.

The main idea of usability testing of this kind of app is that as soon as the user opens the app, the user should get a glance at the market.

a) Exploratory testing

Exploratory Testing is informal testing performed by the testing team. The objective of this testing is to explore the application and look for defects that exist in the application. Testers use the knowledge of the business domain to test the application. Test charters are used to guide the exploratory testing.

b) Cross browser testing

Cross browser testing is testing an application on different browsers, operating systems, mobile devices to see look and feel and performance.

Why do we need cross-browser testing? The answer is different users use different operating systems, different browsers, and different mobile devices. The goal of the company is to get a good user experience regardless of those devices.

Browser stack provides all the versions of all the browsers and all mobile devices to test the application. For learning purposes, it is good to take the free trial given by browser stack for a few days.

c) Accessibility Testing

The aim of Accessibility Testing is to determine whether the software or application is accessible for disabled people or not.

Here, disability means deafness, colour blindness, mentally disabled, blind, old age, and other disabled groups. Various checks are performed, such as font size for visually disabled, colour and contrast for colour blindness, etc.

4) Compatibility testing

This is a testing type in which it validates how software behaves and runs in a different environment, web servers, hardware, and network environment.

8. SCREENSHOTS

8.1 ADMIN PAGE



Login

Your User Name

Your Password

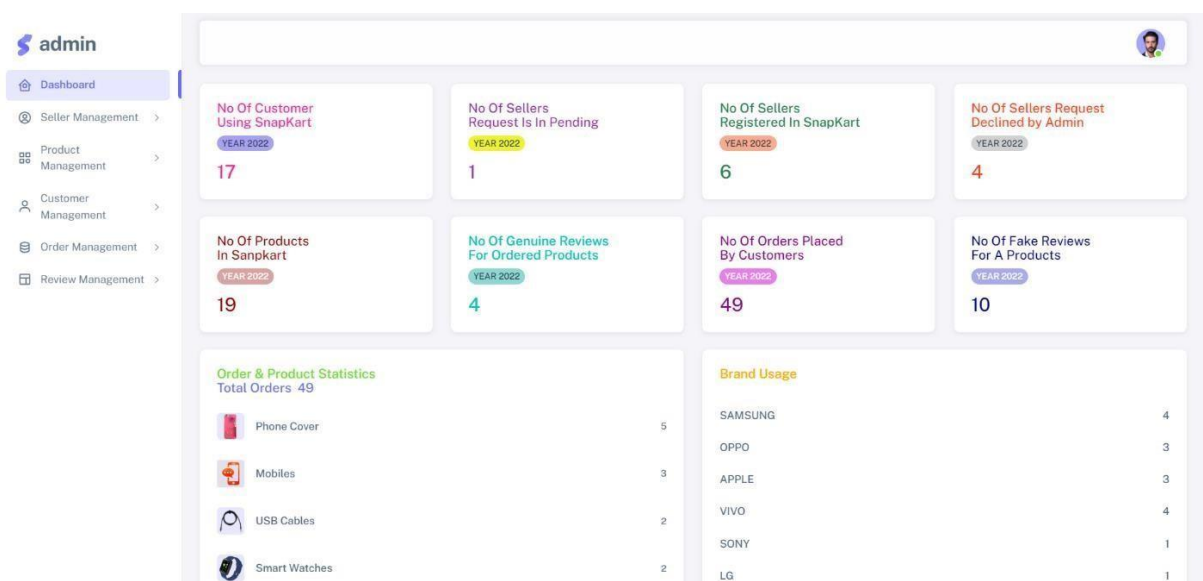
LOGIN

SnapKart

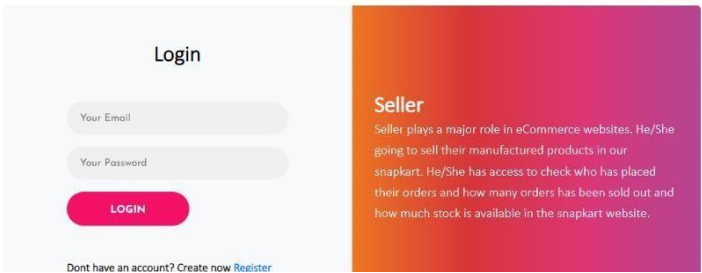
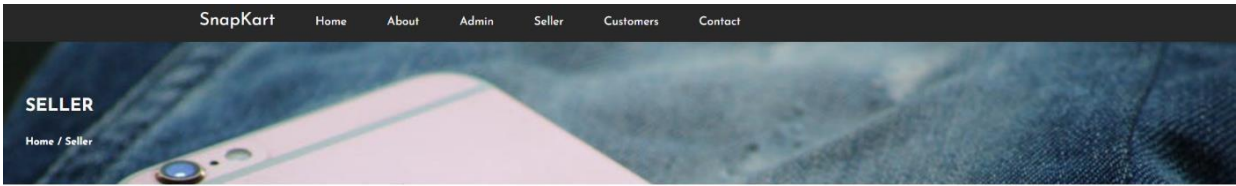
GET IN TOUCH WITH US:

If you have any queries regarding the product quality, delivery status, colors, sizes, models, new launching products, anything you can contact to us.

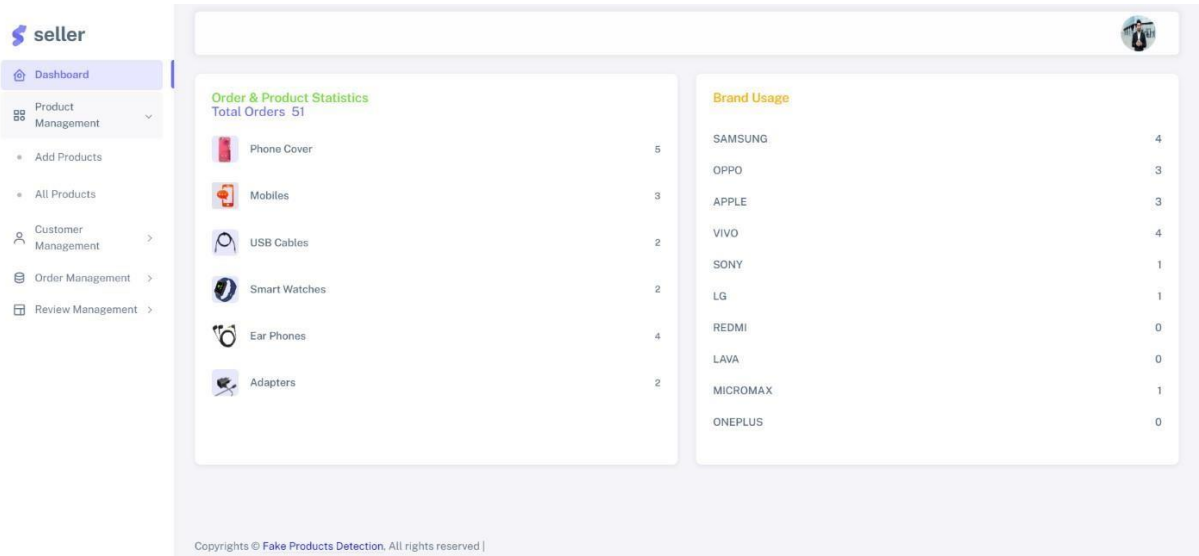
8.2 ADMIN DASHBOARD



8.3 SELLER LOGIN PAGE



8.4 SELLER DASHBOARD



8.5 CUSTOMER LOGIN



Login

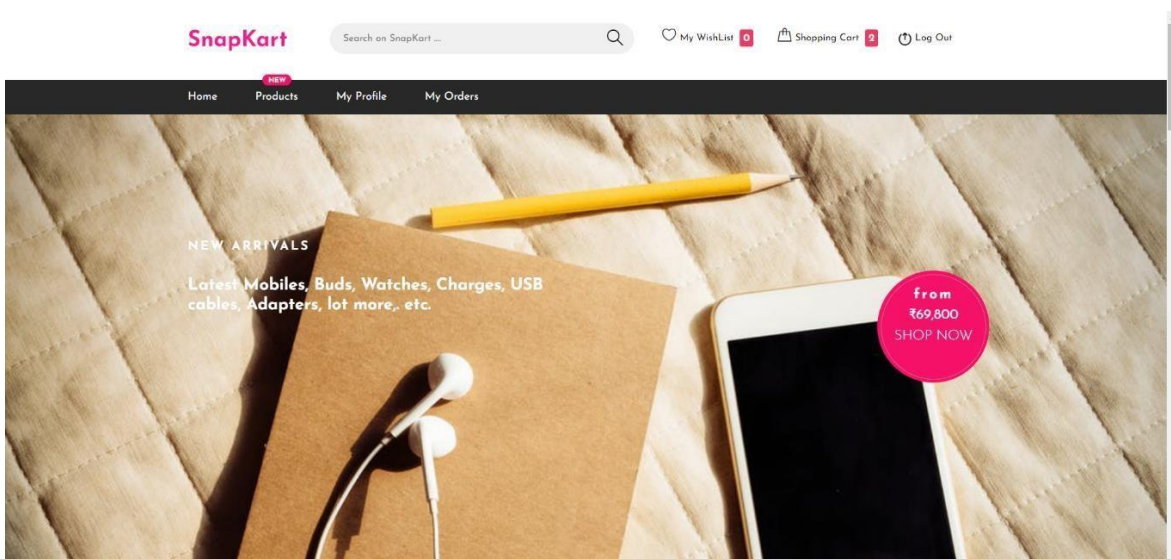
LOGIN

[Dont have an account? Create now Register](#)

Customer

Customers also plays an important role in eCommerce websites. He/She goes to buy the products from the snapkart. He/She has a my cart and my wishlist so that he/she can check in future that how many orders has been placed and how many products are in wishlist.

8.6 CUSTOMER INTERFACE



8.7 PRODUCT REVIEW BY ADMIN

admin Dashboard Seller Management Product Management Customer Management Order Management Review Management		View Product-wise Reviews				
PRODUCT PROFILE	PRODUCT NAME	CUSTOMER NAME	PRODUCT REVIEW	PRODUCT EXPRESSION	PRODUCT SENTIMENT	VIEW GRAPH
	LOXXO® Clear Designed for iPhone 14	harsha vardhan	excellent product	Very Positive		Verify
	LOXXO® Clear Designed for iPhone 14	harsha vardhan	very positive	Positive		Verify
	LOXXO® Clear Designed for iPhone 14	harsha vardhan	worst product ever	Very Negative		Verify
	LOXXO® Clear Designed for iPhone 14	harsha vardhan	some what positiv	Positive		Verify

8.8 SELLER REQUEST ACCEPT BY ADMIN

admin Dashboard Seller Management Product Management Customer Management Order Management Review Management		Seller Request						
SELLER PROFILE	SELLER ID	FULL NAME	CONTACT	EMAIL	AADHAR CARD	GST	ACTIONS	STATUS
	11	Sai	9398884536	annojsaikiran@gmail.com			Accept Decline	ACCEPTED
	10	Sai	9398884536	Saikiran884536@gmail.com			Accept Decline	ACCEPTED
	9	Tanya	8347056544	tanya@gmail.com			Accept Decline	DECLINED
	8	Swarnalatha	7871133322	swarnalatha@gmail.com			Accept Decline	DECLINED
	7	Hema	9955574747	hema@gmail.com			Accept Decline	DECLINED
	6	Malathi	9603422222	malathi@mail.com			Accept Decline	ACCEPTED
	5	Dolly	8800099898	dolly@gmail.com			Accept Decline	ACCEPTED
	4	Krishna	9608584444	krishna@gmail.com			Accept Decline	ACCEPTED

8.9 ADD PRODUCT PAGE

- Dashboard
- Product Management**
- Customer Management
- Order Management
- Review Management

Add Products

PRODUCT NAME

MODEL

BRAND

Select Brand

CATEGORY

Select Category

ORIGINAL PRICE

DISCOUNT PRICE

COMPLETE PHOTO

Choose File

No file chosen

PHOTO2

Choose File

No file chosen

PHOTO3

Choose File

No file chosen

PHOTO4

Choose File

No file chosen

PHOTO5

Choose File

No file chosen

ABOUT

Add Product

8.10 REVIEW MANAGEMENT PAGE

- Dashboard
- Product Management
- Customer Management
- Order Management
- Review Management**

View Product-wise Reviews

PRODUCT PROFILE	PRODUCT NAME	CUSTOMER NAME	PRODUCT REVIEW	EXPRESSION	SENTIMENT
	LOXXO® Clear Designed for iPhone 14	harsha vardhan	excellent product	Very Positive	
	LOXXO® Clear Designed for iPhone 14	harsha vardhan	very positive	Positive	
	LOXXO® Clear Designed for iPhone 14	harsha vardhan	worst product ever	Very Negative	
	LOXXO® Clear Designed for iPhone 14	harsha vardhan	some what positiv	Positive	

8.11 SELLER ACCEPT THE ORDER

- Dashboard
- Product Management
- Customer Management
- Order Management**
- Review Management

View Orders

PRODUCT NAME	QUANTITY & PRICE	TOTAL PRICE	SHIPPING ADDRESS	ORDER DATE	ORDER STATUS
JBL T460BT by Harman, Wireless On Ear Headphones	1 x 2199	2199	('Mansoorabad Road, Plot no 211, Mansoorabad, LB Nagar')	March 24, 2024, 6:51 a.m.	<button>Accept</button> <button>Reject</button>
LOXXO Clear Designed for iPhone 14	1 x 299	299	('golnaka road, plot no 25, nitresh nagar')	March 21, 2024, 5:53 a.m.	<button>Accept</button> <button>Reject</button>
SUANPOT for LG Wing 5G with RFID Blocking Leather Wallet case	1 x 2200	2200	('golnaka road, plot no 25, nitresh nagar')	March 21, 2024, 5:53 a.m.	<button>Accept</button> <button>Reject</button>

8.12 ORDER MANAGEMENT

- Dashboard
- Product Management
- Customer Management
- Order Management**
- Review Management

View Orders

SELLER NAME	CUSTOMER NAME	PRODUCT PROFILE	PRODUCT NAME	QUANTITY & PRICE	TOTAL PRICE	SHIPPING ADDRESS
Aravind	Satya		JBL T460BT by Harman, Wireless On Ear Headphones	1 x 2199	2199	('golnaka road, plot no 25, nitresh nagar')
Aravind	Satya		LOXXO Clear Designed for iPhone 14	1 x 299	299	('golnaka road, plot no 25, nitresh nagar')
Aravind	Satya		SUANPOT for LG Wing 5G with RFID Blocking Leather Wallet case	1 x 2200	2200	('golnaka road, plot no 25, nitresh nagar')
Aravind	harsha vardhan		aravind	1 x 85	85	('tarnaka')

8.13 CERTIFIED REVIEW BY USER

All Review for this Product

**Satya**
it is good but it will work only 30 days

Certified Buyer 

March 21, 2024, 5:54 a.m.

**Surya**
bad

Certified Buyer 

March 24, 2024, 6:43 a.m.

8.14 UNCERTIFIED REVIEW BY USER

All Review for this Product

**Uday kiran**
bad

Uncertified Buyer 

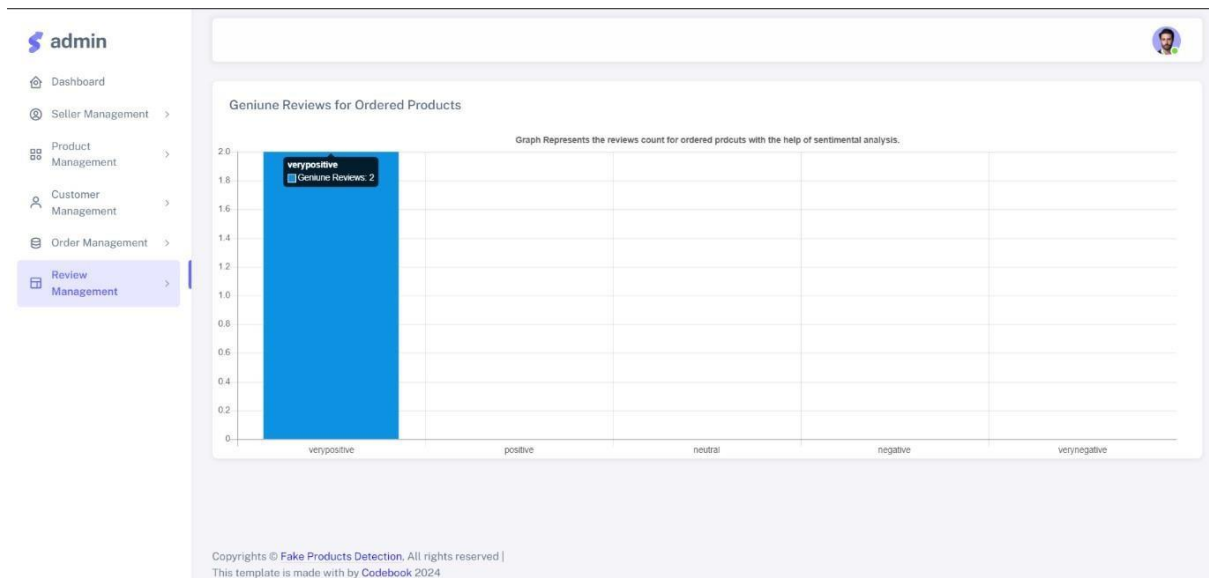
March 21, 2024, 5:01 a.m.

**Surya**
good

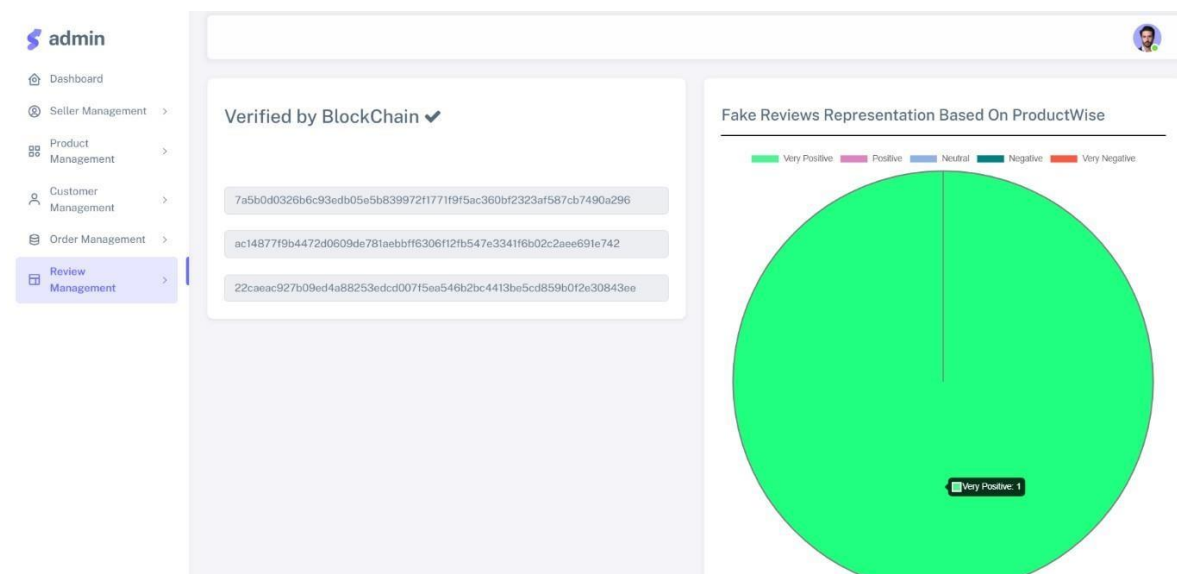
Uncertified Buyer 

March 24, 2024, 6:44 a.m.

8.15 REVIEW FOR ORDERED PRODUCT



8.16 AFTER VERIFY QR CODE WILL GENERATE



8.17 ORDER LIST

Orders List History				
Product Profile	Product Name	status	Review	scan
	JBL T460BT by Harman, Wireless On Ear Headphones 	placed	Review	

8.18 QR CODE GENERATE FOR VERIFIED PRODUCT

Orders List History				
Product Profile	Product Name	status	Review	scan
	JBL T460BT by Harman, Wireless On Ear Headphones 	order accepted	Review	

9. CONCLUSION

With the large volume of counterfeit items available online, counterfeit products are expanding quickly. There is a pressing need to detect counterfeit goods, and blockchain technology is being used to accomplish this goal. Additionally, the data is encoded as a QR code. Customers or consumers may discover counterfeit products by scanning the QR code. In blockchain technology, digital information on items may be kept in the form of blocks. Firebase cloud storage may be used to store the data. Thus, the suggested approach benefits the client by detecting counterfeit items across the supply chain. It is possible to access information about the product's history and present owner by scanning the QR codes that have been affixed to it.

10. REFERENCES

- [1] Si Chen, Rui Shi, Ren, Jiaqi Yan, Yani Shi, "A Blockchain-based Supply Chain Quality Management Framework", 14th, IEEE International Conference on e-Business Engineering, 2017.
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